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Review

Pulsed Field Gradient Nuclear Magnetic Resonance and Diffusion Analysis in Battery Research

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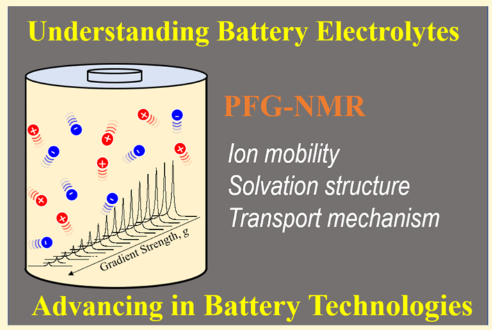
ABSTRACT: Pulsed-field gradient nuclear magnetic resonance (PFG-NMR) is a widely used method for determining the diffusion coefficient of ions and molecules both in the bulk and when confined (e.g., within porous materials). Due to the nature of diffusion phenomena and the correlation of these processes with the structures of isolated molecules or clusters, studies of diffusion can be used to extract both dynamic and structural information from complex mixtures, including battery electrolytes composed of cations, anions, and solvent molecules. PFG-NMR presents a powerful opportunity for battery scientists to quantify electrolyte properties, such as time scales for dynamics, transference numbers, and solvation structures of active ions that vary due to ion−ion and ion−solvent interactions. These measurements and the derived information about molecular interactions can ultimately be correlated with real battery performance. The purpose of this review is to provide readers with an overview of the basic principles and experimental considerations when undertaking PFG-NMR for battery electrolyte research. In this review, we will first (1) introduce basic PFG-NMR experiments, parameters, and the proper setup for acquiring accurate diffusion coefficients and (2) discuss artifacts that can arise in diffusion measurements, including their diagnosis and suppression. Second, we show the ultimate power of careful analyses of diffusion coefficients for extracting dynamic and structural properties of a wide range of electrolyte types (i.e., dilute, concentrated, polymer, and solid-state) through a review of selected literature. In addition, other NMR methods are briefly introduced, including relaxation measurements and Overhauser dynamic nuclear polarization (ODNP) NMR.



■ INTRODUCTION

From very early on in the advent of nuclear magnetic resonance (NMR) as a spectroscopic technique, it was demonstrated that it is possible to measure the self-diffusion coefficient of a mobile species by coupling the spin−echo method with an accompanying constant magnetic field gradient during the course of the pulse train.[1] Effectively, this produces attenuation in the measured signal because the field gradient makes the Larmor frequencies of the nuclear spins under resonance a function of their position in the sample volume, so if the spins are undergoing motion relative to this field gradient, the echo will not refocus completely. But in order for this original method to be rigorously applied, the system must satisfy certain experimental criteria and conditions, such as tailoring the radio frequency (rf) field ($\mathbf{H}_1$) amplitude and gradient strength to properly capture the diffusivity, and the spin system must manifest a sufficiently long spin−spin relaxation time (T_2) for the observed nucleus. A decade later, the well-known and now ubiquitous Stejskal−Tanner method debuted,[2] which utilized a short-duration field gradient pulse, and therefore this method is known as PFG-NMR. This development extended the usability of the spin−echo approach for diffusion measurement via magnetic resonance by mitigating many of the restrictions of the original, constant-gradient spin−echo method. In the decades since this seminal development, this simple spin−echo PFG-NMR sequence has been modified and improved by ever-increasing advances in sophistication, such as the switch to predominantly stimulated-echo-based sequences, the incorporation of bipolar gradients, and the use of additional rf and gradient pulses to carefully design the experiment to suit a broadening range of applications.[3] Many of these modified PFG-NMR sequences were essentially developed for the suppression of various types of interference or artifacts that can have significant impact on the measured value of the diffusion coefficient. These artifacts can stem from both the measuring system itself[4] and the samples, with common

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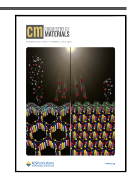

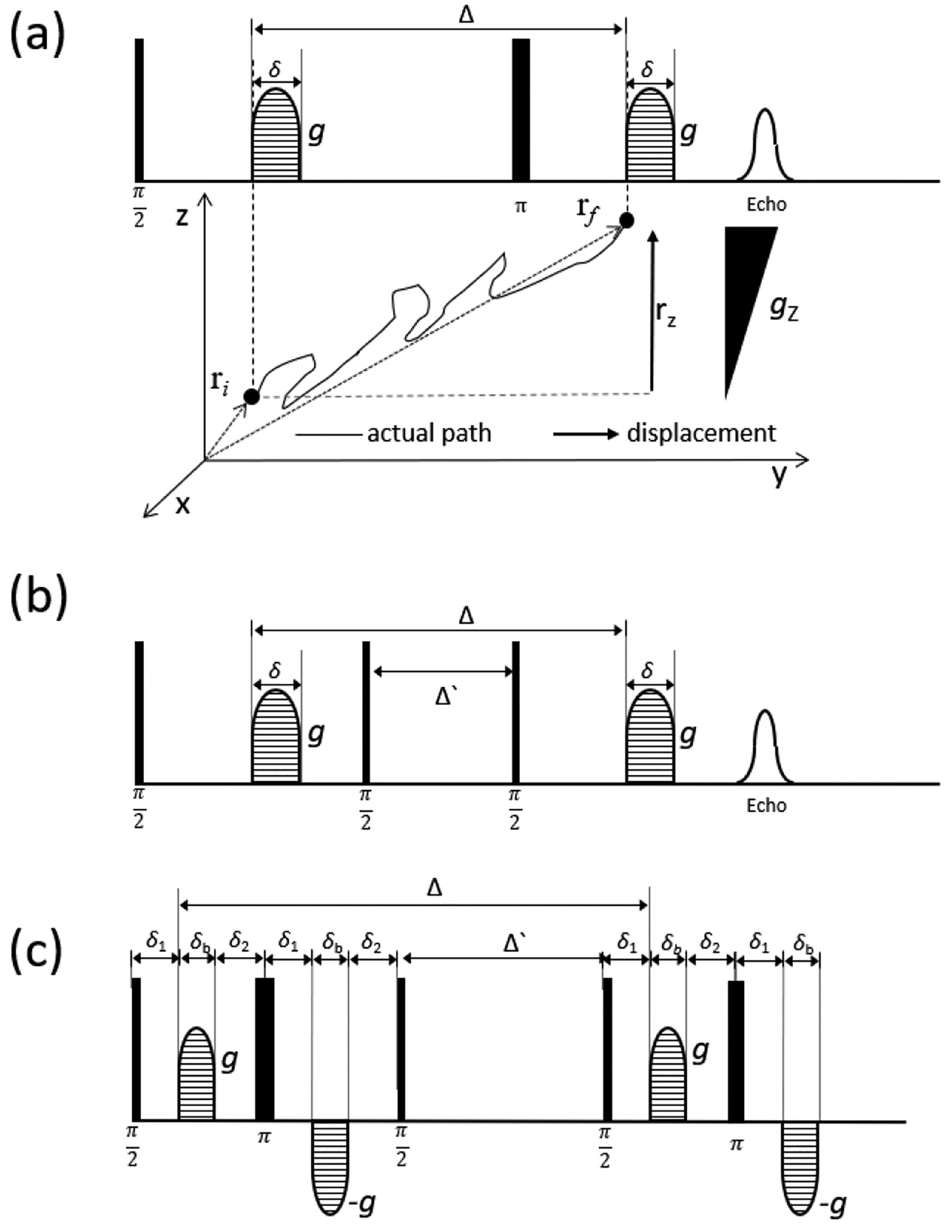

Figure 1. Representative sequences for PFG-NMR: (a) PFG spin−echo (SE) and schematic outline for the diffusion measurement using the SE sequence, illustrating that PFG-NMR measures the displacement ($\mathbf{r}_z$) rather than an actual diffusion length during the measurement period (Δ). The first and second gradient pulses encode and decode the initial ($\mathbf{r}_i$) and final ($\mathbf{r}_f$) positions of the spins in motion. Commonly, a z-gradient (g_z) is used for the diffusion measurement in most commercial NMR spectrometers, in which case the gradient strength g is along z-axis ($g = g_z$). (b) Stimulated echo pulse sequence with monopolar gradients (STE_MPG). (c) Stimulated echo pulse sequence with bipolar gradients (STE_BPG). In the STE_BPG, the gradient length is $\delta = 2\delta_b$.

examples being convection in bulk liquid samples,[5] internal background gradients (g_0) in viscous liquids,[6] and fluid confinement in porous materials.[7]

While the PFG-NMR technique has already been widely used for liquids in which the spins possess long relaxation times and undergo relatively fast diffusion processes, its applications to solid state materials and fluid confined to porous materials are more challenging due to the limitations of the gradient strength and the comparatively shorter relaxation times inherent to these types of samples. There have, of course, been several excellent PFG-NMR review articles and book chapters published dealing with the basic theory, experimental methods, suppression of artifacts in the diffusion measurement, and diffusion-ordered spectroscopy (DOSY; which couples PFG-NMR with conventional NMR spectroscopy to separate similarly diffusing species in a mixture based on their resonance frequency, and vice versa).[3,5,8−15] However, for a researcher who is not familiar with PFG-NMR, despite possibly being an NMR expert, it is not straightforward to figure out the limitations of PFG-NMR applications and how the experimental parameters and artifacts influence the PFG-echo profile (i.e., the spectroscopic signature of the diffusion coefficient) in actual PFG-NMR measurements. Therefore, this article will be focused first on understanding (1) basic PFG-NMR parameters and descriptions of common pulse sequences, (2) approaches to setting up experimental

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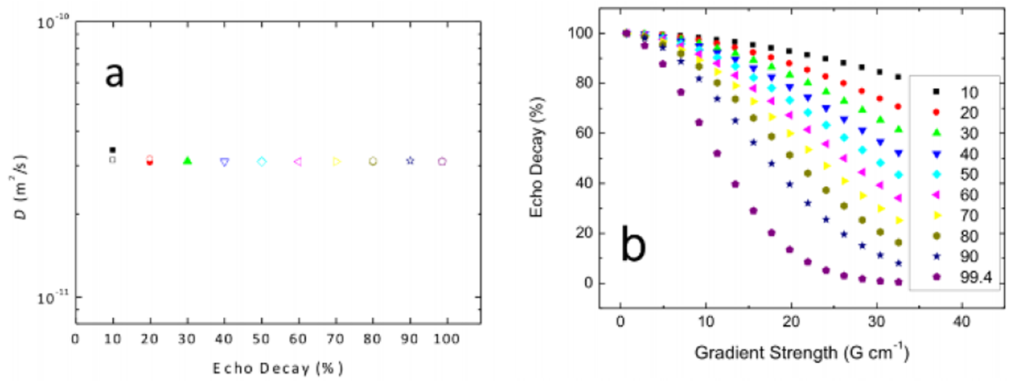

Figure 2. (a) Diffusion coefficients as a function of the magnitude of the decay in the PFG echo, and (b) corresponding echo profiles obtained using ^{1}H PFG-NMR with two different ^{1}H resonances of 1-butyl-3-methylimidazolium bis(trifluoromethlysulfone)imide [C_4mim][TFSI] ionic liquid.[16] Reprinted with permission from K. S. Han et al. *J. Phys. Chem. C* **2013**, *117*, 15754−15762 (ref 16). Copyright 2013 American Chemical Society.

parameters, and (3) the behavior of apparent diffusion coefficients due to the presence of artifacts, such as sample convection and background gradients (g_0), as well as methods for artifact suppression. Second, it will be shown how to analyze measurements of the diffusion coefficient in the specific context of understanding electrolyte systems at the molecular scale, as an example of an application where PFG-NMR is finding increasingly prevalent use.

THE PFG-NMR EXPERIMENT

PFG-NMR Parameter Basics. The PFG-echo intensity obtained after performing a basic PFG-NMR pulse sequence (Figure 1) is a function of the parameters γ, δ, g, and Δ and is related to the self-diffusion coefficient, D, by the Stejskal−Tanner equation,[2]

$$S(g) = S(0)\exp\left[-D(\gamma\delta g)^2\left(\Delta - \frac{\delta}{3}\right)\right] = S(0)\exp[-Db] \tag{1}$$

where $S(g)$ and $S(0)$ are the intensity of the PFG-echo profile at the gradient strength of g and 0, respectively, γ is the gyromagnetic ratio of the observed nucleus, δ is the gradient duration (in tandem, δg is the "area" of the gradient in the pulse sequence), and Δ is the distance between the two gradient pulses (i.e., the diffusion time). The equation indicates that the PFG-echo intensity becomes smaller as the parameters γ, δ, g, and Δ are increased in magnitude. Similarly, the intensity also decreases with larger D (i.e., faster diffusion). In modern PFG-NMR sequence design, the PFG-echo profile, which must be fit to eq 1 to extract the diffusion coefficient, is most commonly obtained as a function of gradient strength, while the remaining parameters, δ and Δ, are fixed (with γ obviously being intrinsic to the nucleus). Nevertheless, the selection of values for δ and Δ plays a critical role in matching the available gradient range of the spectrometer to the magnitude of the diffusion process being observed. Additionally, the dependence of eq 1 on γ means that it is highly preferable to observe the high-γ nuclear constituents of a given diffusing species for the most efficient experimental acquisition. In measuring very fast diffusion processes, like self-diffusion in low-viscosity liquids, there are very few restrictions to how these parameters can be configured. For example, it is possible to obtain a full decay of the echo profile with a small gradient area (δg) and a long diffusion delay (Δ), on account of the relatively lengthy relaxation times, T_1 and T_2, in this motional regime (under the extreme narrowing condition given by $\omega_0\tau_c \ll 1$, where ω_0 is the Larmor frequency and τ_c is the rotational correlation time, $T_2 \approx T_1$). As the self-diffusion coefficient decreases, it is necessary to correspondingly increase the gradient strength to obtain a similarly full decay in the echo profile. This highlights the importance of the maximum gradient strength that can be generated by the diffusion probe, especially for applications involving slower diffusion. In the fast-motion regime ($D \sim 10^{-9}$ m^2/s, e.g., self-diffusion of water at room temperature), gradient strengths of only a few tens of Gauss per centimeter (G/cm = 10^{-2} Tesla per meter, T/m) are adequate, whereas slower diffusion ($D < 10^{-11}$ m^2/s) in viscous liquids or of ions in solid-state ion conductors will typically require gradient strengths greater than 2000 G/cm. However, if it is not possible to obtain the full decay of the echo profile solely from increasing the gradient strength, due to instrumental limitations on the maximum achievable gradient strength, a longer diffusion delay can be employed, provided that the signal remains observable when factoring in relaxation. Nevertheless, when the maximum gradient strength and the longest feasible diffusion delay are still not sufficient to generate full decay in the echo profile, the correct diffusion coefficient can be obtained from the echo profile with a 20% decay in some cases,[16] as shown in Figure 2.

As this discussion illustrates, there is an upper limit to the feasibility of PFG-NMR measurements enforced by the joint considerations of maximum available gradient strength and sample relaxation, which translates to a floor on the minimum rate of diffusion that can be reliably measured. Typically, nuclear relaxation becomes faster when the molecules undergo slower translational and rotational motions. While T_1 relaxation is still longer in the so-called "strong collision limit", where $\omega_0\tau_c \gg 1$, the shorter T_2 relaxation associated with this motional regime renders it difficult to still observe any signal once the time necessary to apply a PFG pulse sequence comprised of multiple rf and gradient pulses and intervening delays has elapsed. Here, it is worthwhile to discuss the PFG-NMR pulse sequences in relation to the T_1 and T_2 relaxation times. The representative PFG-NMR sequences are presented in Figure 1: the spin−echo (SE) and stimulated echo (STE) with monopolar (STE_MPG) and bipolar gradients pulses (STE_BPG). Supposing that there are no gradient pulses, the echo intensity is a function of T_2 relaxation in the SE sequence

(Figure 1a), whereas for STE sequences, it depends mainly on T_1 relaxation during the diffusion time (Δ) because the second and third rf pulses in Figure 1b,c, respectively, rotate the magnetization to the z-axis (known as "z-storage"), where only longitudinal relaxation of the spin system takes place during the time interval Δ'. However, there is still transverse relaxation before (and after) the time interval Δ', so the role of T_2 is not completely eliminated. As a result, it is generally necessary to have at least several milliseconds of T_2 to perform PFG-NMR experiments. The STE sequences have an advantage because $T_1 \geq T_2$ in many cases, but the trade-off can be relatively weak peak intensity, which is reduced by a factor of 2 compared with the SE sequence because of losses inherent to the stimulated echo formation.[3] This relaxation-based limitation can be a significant complication in diffusion measurements of systems such as ions/molecules in viscous liquids and gels, and of species residing on the surface of porous materials, and is compounded by the slower diffusion typically occurring in such systems.

PFG-NMR Parameter Setup. In this section, the setting of PFG-NMR parameters will be discussed, in terms of empirical aspects. DOSY and diffusion coefficient measurements can be performed with ordinary liquid NMR spectrometers that are equipped with a solution-state NMR probe that can generate a z-gradient. Often, an ordinary commercial solution-state NMR probe can generate a z-gradient of a few tens of Gauss per centimeter (G/cm)[16] while some commercial imaging probes with a gradient of a few hundreds of Gauss per centimeter are available for 3-dimensional diffusion measurements.[17,18] It is important to empirically verify the calibration of the gradient strength, to confirm that the current pulses supplied by the gradient amplifier are indeed producing what is expected by the nominal input values for gradient strength. There are several means of accomplishing this, with ^{1}H PFG-NMR of pure water widely used. If the calibrated gradient strength is correct, the water diffusion coefficient will be 2.299×10^{-9} m^2/s at 25 °C.[19] Since diffusion is a thermally activated process and is highly sensitive to temperature, calibration of the temperature regulation system should first be performed, and the temperature regulation must be in use during the gradient calibration process. Many common temperature calibration methods for NMR experiments employ chemical shift-based thermometers, with, e.g., ethylene glycol or methanol,[20] and these are typically well-suited to PFG-NMR setup as well. It is also important to note that, particularly on higher-field systems, radiation damping can significantly impact ^{1}H measurements on proton-dense systems, so calibrating on the residual proton signal in D_2O or in "doped water" with minute additions of $CuSO_4$ or $MnCl_2$ is preferable.[19] It is also highly preferable to use a pulse sequence with bipolar gradient pulses (e.g., STE_BPG represented in Figure 1c) for this purpose, due to the unique ability of this type of sequence to suppress artifacts owing to a residual B_0 gradient and gradient pulse mismatch.[4]

Having established the robustness of the gradients and thermal regulation and suitably calibrated these systems, the task now turns to effectively navigating the (Δ, δ, g) parameter space that underpins all PFG-NMR experiments. Basically, as shown in Figure 3, for 3-dimensional free diffusion of the type generally observed in bulk liquids, the measured diffusion coefficient will be independent of the diffusion delay (Δ). In contrast, restricted diffusion of ions/molecules confined in porous media is a function of diffusion delay as a consequence of the presence of diffusion barriers. Hence, the pore structure can be probed from the diffusion delay dependence of the derived diffusion coefficients.[21] Empirically, the most preferable path is to choose the diffusion delay (Δ), followed by gradient pulse length (δ), and then subsequently array the gradient strength (g) to acquire the echo profile. A straightforward method to estimate the maximum gradient strength required to obtain full decay ($\geq$90%) of the echo profile is to simply compare the peak intensity between the two echo spectra obtained using a weak and stronger gradient strength (with the former perhaps 10% of the strength of the latter). Then, if it is necessary, it may be possible to change the parameters (Δ, δ, and/or g) to better optimize the maximum attenuation attained in the echo profile. This process is a particularly time-efficient means of finding the optimal (Δ, δ, g) parameters. After finishing set up of the PFG-NMR parameters, the PFG-echo profile can be accumulated with an array of gradient strengths varied in ~15 equal steps, with the maximum and minimum gradient strength determined in the previous procedure. The experimental situation may call for a different number of increments in the gradient strength array, depending on the signal-to-noise ratio (and therefore the scatter in the attenuation curve), the degree of attenuation that can be achieved, or the particular diffusion phenomenon under investigation. For example, the "NMR diffraction"[22] observed for an ionic liquid crystal in the ^{7}Li PFG-echo profile obtained with 30 gradient steps was not clearly shown in the echo profile obtained with 15 gradient steps.[23] While almost all bulk liquids give the same diffusion coefficient regardless of the length of diffusion delay (Δ), it is sound methodology to confirm the lack of Δ-dependency of the diffusion coefficient at an early stage of the experimental setup as a precaution. As an example, as shown in Figure 3, Δ-dependent (restricted) diffusion behavior was unexpectedly observed from a ternary electrolyte solvent mixture (comprised of ethylene carbonate, propylene carbonate, and ethyl-methyl carbonate; EC/PC/EMC) due to the presence of a solid/liquid binary phase.[24]

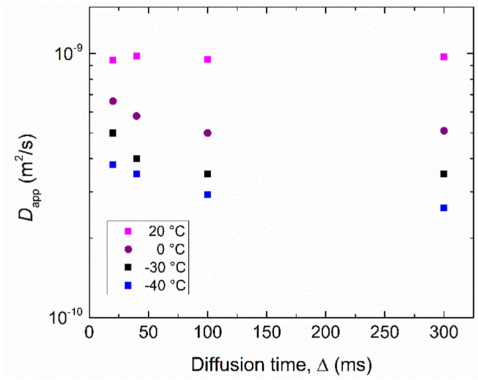

Figure 3. Temperature-dependent apparent diffusion coefficients (D_{app}) obtained as a function of diffusion delay (Δ) from the solvent mixture of EC/PC/EMC (1:4:5 by weight). At lower temperatures, typical restricted diffusion behavior developed due to the presence of a solid/liquid binary phase, indicated by the initial reduction in D_{app} with increasing Δ.[24] Reprinted with permission from K. S. Han et al. *J. Chem. Phys.* **2014**, *141*, 104509 (ref 24). Copyright 2014 AIP Publishing.

To illustrate this parameter selection process, ^{25}Mg PFG-echo decay profiles simulated using the Stejskal–Tanner equation (eq 1) as a function of given diffusion coefficients with fixed Δ and δ values are presented in Figure 4. Here, the

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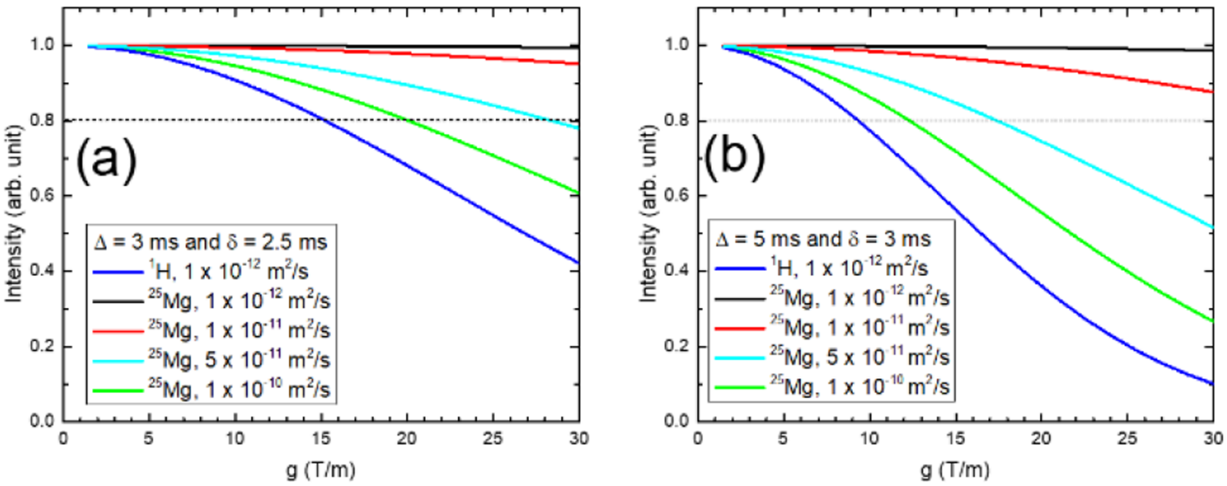

Figure 4. Simulated ^{25}Mg PFG-echo decays as a function of diffusion coefficient, with fixed values of (a) Δ = 3 ms and δ = 2.5 ms and (b) Δ = 5 ms and δ = 3 ms, using the Stejskal−Tanner equation. Both plots also contain a ^{1}H PFG-echo decay simulated at diffusion coefficient $D = 1 \times 10^{-12}$ m^2/s. Horizontal dotted lines show the 20% echo decay.

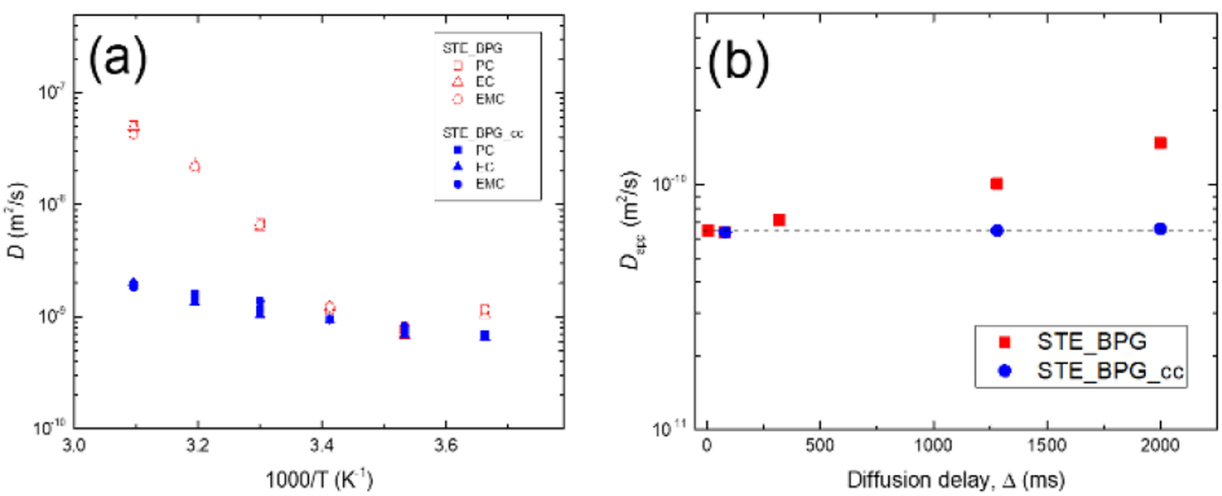

Figure 5. (a) Temperature-dependent diffusion coefficients obtained using STE_BPG and the pulse sequence developed for the convection compensation (STE_BPG_cc) from the solvent mixture of EC/PC/EMC (1:4:5 by weight)[24] and (b) diffusion delay dependent D_{app} of the TFSI$^-$ anion in a 0.75 M Mg(TFSI)$_2$/DME solution obtained with ^{19}F PFG-NMR at −5 °C. Due to the presence of sample convection, the D_{app} increased with the increase of diffusion delay with the STE_BPG sequence, whereas it is completely suppressed with the STE_BPG_cc method (as evidenced by the constant D_{app} for all Δvalues). Reprinted with permission from K. S. Han et al. *J. Chem. Phys.* **2014**, *141*, 104509 (ref 24). Copyright 2014 AIP Publishing.

Δ values are fixed at 3 or 5 ms because it may not be possible to detect the signal with longer values than 2−3 ms in actual systems due to the fast ^{25}Mg nuclear relaxation (typically on the order of only several milliseconds). As can be seen by comparison between Figure 4a and Figure 4b, the echo decay is enhanced by the increase of the Δ and δ values and also higher diffusion coefficients. Conversely, while the diffusion coefficients are the same, the ^{1}H PFG-echo decay plotted alongside is much larger than the ^{25}Mg PFG-echo decay owing to the appreciable difference in gyromagnetic ratios between these two nuclei: $\gamma(^1\mathrm{H}) = 2\pi\cdot 42.577$ rad·MHz/T $\gg \gamma(^{25}\mathrm{Mg}) = 2\pi\cdot 2.606$ rad·MHz/T. This simple simulation demonstrates the correspondence between the PFG-echo decay and the PFG-NMR parameters and illustrates how they interact with the diffusion coefficient of the targeted species in the sample. The benefit of higher gradient strength in the diffusion measurement is also evident, as reasonable attenuation in these echo profiles for most cases is only realized with a maximum gradient strength around 30 T/m, at the upper limit of most commercially available gradient probes. Considering both the restrictions on the PFG-NMR delays imposed by relaxation and the significant gradient strength mandated by the low gyromagnetic ratio, these simulations also demonstrate why it is nontrivial—indeed sometimes practically impossible—to apply PFG-NMR to quadrupolar nuclei such as ^{25}Mg. Nevertheless, recent success in performing ^{25}Mg PFG-NMR was achieved with aqueous Mg([CF$_3$SO$_2$)$_2$N])$_2$ electrolytes. The results showed that a Mg^{2+} ion solvated by six water molecules forms a symmetrical rigid octahedral configuration, such that the Mg^{2+} ions exist as $[Mg(H_2O)_6]^{2+}$ cations, resulting in the similar hydrodynamic radius for Mg ions and TFSI$^-$ anions. It was possible to obtain a ^{25}Mg diffusion measurement in this case due to the slower nuclear relaxation,

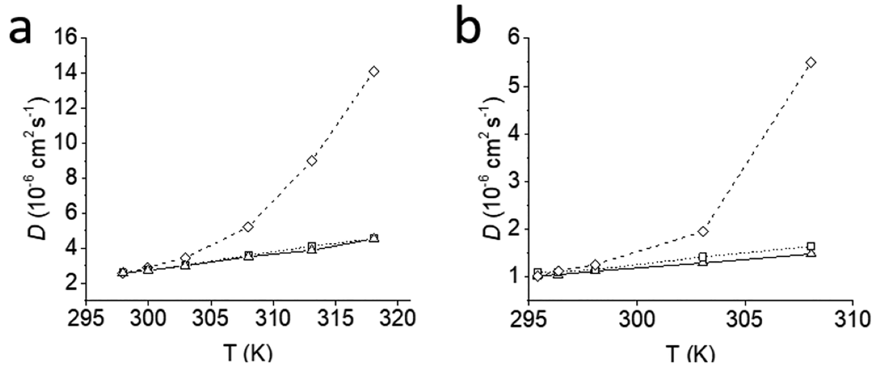

Figure 6. Temperature dependence of the diffusion coefficients of (a) β-cyclodextrin in D_2O and (b) lysozyme in H_2O/D_2O with and without sample spinning. Three different data are represented: (◇) BPLED experiment without rotation, (□) BPLED experiment with rotation, and (△) compensated-BPLED sequence.[26] BPLED and compensated-BPLED have the STE_BPG and STE_BPG_cc sequences, respectively, as their core, but are equipped with an added pair of rf pulses at the end of the pulse train for allowing a longitudinal eddy current delay (LED)[3] before aquiring the signal. Adopted with permission from N. Esturau et al. *J. Magn. Reson.* **2001**, *153*, 48−55 (ref 26). Copyright 2001 Elsevier.

especially the spin−spin (T_2) relaxation, which is a result of the symmetrical solvation of the Mg^{2+} ions.[25]

Artifacts due to Sample Convection and Methods for Suppression. As discussed above, active temperature regulation is often necessary because of the high thermal sensitivity of diffusion coefficients. It is also necessary for measurement of temperature-dependent diffusion processes and for the extraction of activation energies for the translational motion of the diffusing species. Due to the configuration of the NMR probe, sample temperature control is typically accomplished with nitrogen gas flowing around the base of the sample. Therefore, there can be a temperature gradient built up along the z-axis that causes convection in the sample, artificially inflating the diffusion coefficient as shown in Figure 5. As mentioned previously, the diffusion coefficient should be independent of the diffusion delay for a bulk liquid undergoing 3-dimensional, unrestricted diffusion. However, under the influence of sample convection, the measured value will increase with increasing diffusion delay, as shown in Figure 5b. It is worthwhile to emphasize that the convection artifact, which is not severe in viscous liquids, can nevertheless still be significant in low-viscosity liquids at low temperature (which are in a nominally more viscous regime), such is the extent of the influence of the potential temperature gradients. As shown in Figure 5b, the simplest way for the test of the presence of an impactful sample convection artifact is by repeating the measurements with increased diffusion delay.

As demonstrated in Figure 5, the diffusion coefficients obtained with STE_BPG are higher than those obtained with the PFG sequence developed for convection compensation (STE_BPG_cc)[5] in temperature-controlled experiments, at both higher and lower temperatures. The deviation due to the convection became larger as the amount of increase or decrease in the temperature from ambient temperature became larger in magnitude. To counter this, the STE_BPG_cc sequence for convection compensation is constructed around a double echo, borrowing the idea that the first moment of the field gradient across the sample is nulled by even numbers of echoes in CPMG sequences.[15] However, this style of "double PFG" experiment accordingly necessitates higher numbers of rf and gradient pulses, with associated spacing delays, compared with SE and STE_BPG sequences. A close examination of Figure 5b reveals that the minimum diffusion delay that was used for STE_BPG is not available for STE_BPG_cc, due to the additional sequence time arising from these additional convection compensation rf and gradient pulses. Notably, there may be insufficient time for species with rapid relaxation to encode the entire "double PFG" sequence prior to complete signal decay with the minimal requirements for spacing delays between rf and gradient pulses, along with the finite duration of each of these events. It is also important to be wary of the potential impact that additional rf and gradient pulses can have in terms of signal distortion, owing to eddy currents induced by the extra gradient pulses and the potential for rf heating of the sample, which will be highly dependent on the probe.

In such cases, possible workarounds include rotating the sample (Figure 6),[26] using a special NMR tube with a short spatial extent along the z-axis for the sample, and susceptibility approximately matched to the sample above and below (i.e., so-called "Shigemi tubes" for aqueous samples)[27] or placing the sample in capillary tubes themselves contained in the NMR tube (to limit the diameter for flow),[28,29] as shown in Figure 7. However, Shigemi tubes and NMR tubes with capillary inserts sacrifice the usable active volume for the sample (and, hence, sensitivity) and spectral resolution, due to the adverse impacts of such configurations for magnet shimming (which is the optimization of magnet field homogeneity). Also, it is important to locate short samples, such as Shigemi tubes, proximal to the center of the gradient to avoid a slower attenuation at the edge of the gradient. If amenable to the sample, a brief one-dimensional, frequency-encoding MRI experiment can assist with this alignment process.[30] Due to these disadvantages, using Shigemi tubes or the capillary inserts may not be a suitable choice for the determination of diffusion coefficients from multiple ions/molecules simultaneously, especially in ^1H PFG-NMR where the chemical shift separation between species may be quite small and the lower signal intensity from dilute components may be difficult to separate from solvent signals. As an example, Figure 8 shows the ^1H NMR spectrum of 0.85 M *N*-(ferrocenylmethyl)-*N*,*N*-dimethyl-*N*-ethylammonium bis(trifluoromethane)-sulfoneimide, [Fc1N112][TFSI], dissolved in EC/PC/EMC (4:1:5 wt %) obtained with a diffusion probe. This clearly shows that it may not be possible to determine the diffusion coefficients of each solvent molecule precisely if the magnetic field homogeneity were to degrade even slightly. To minimize these disadvantages, the simplest approach we tested is using an ordinary 5 mm outer diameter (OD) NMR tube containing

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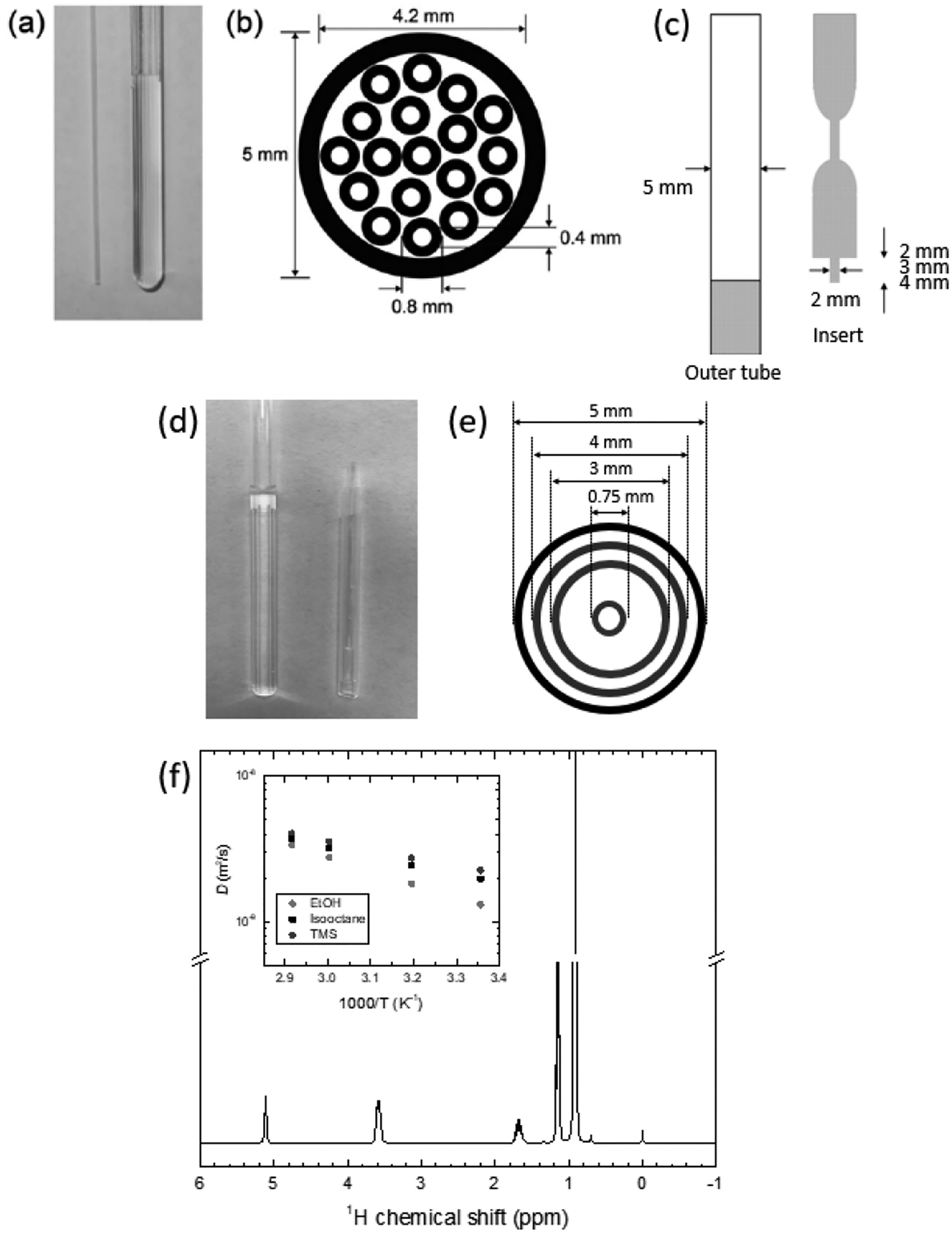

Figure 7. (a) Photograph of a 0.8 mm OD glass capillary (left) and a 5 mm NMR tube containing 18 capillaries (right). (b) Cross-sectional diagram of a 5 mm NMR tube containing 18 capillaries. The glass areas across which the sample solution cannot move are shown in black.[28] Reprinted with permission from T. Iwashita et al. *Magn. Reson. Chem.* **2016**, *54*, 729−733 (ref 28). Copyright 2016 John Wiley and Sons, Ltd. (c) Modified NMR sample tube giving an annular sample volume supplied by Shigemi, Tokyo.[27] Reprinted with permission from K. Hayamizu et al. *J. Magn. Reson.* **2004**, *167*, 328−333 (ref 27). Copyright 2004 Elsevier Inc. (d) Photograph of a 5 mm NMR tube (left) containing the three-layered insert (right) composed of a 1.75 mm OD glass capillary, along with 3 and 4 mm OD commercial NMR tubes. (e) Cross-sectional diagram of a 5 mm NMR tube containing the three-layered insert. If it is necessary to lock the signal, then a deuterated solvent can be loaded into the 1.75 mm OD glass capillary and sealed. (f) ^{1}H spectrum extracted from the first slice of the PFG-echo profile obtained using STE_BPG with the NMR tube containing the insets represented in (d) and (e). The inset provides the temperature-dependent diffusion coefficients of ethanol, isooctane, and TMS for the sample, which contains a 20% volume of ethanol mixed with isooctane and containing 1% volume of TMS.

a three-layered set of inserts, which is composed of a capillary tube and ordinary 3 and 4 mm OD NMR tubes as represented in Figure 7d,e. The preliminary test was performed at 50 °C with a mixture of 1,3-dioxolane (DOL) and 1,2-dimethoxyethane (DME) (1:1, v/v), which has a viscosity of ≤0.588 mPa·s at 25 °C.[31,32] We confirmed that the diffusion coefficients obtained using STE_BPG with inserts and STE_BGP_cc without inserts are the same, and both are also 100 times smaller than the diffusion coefficient obtained using STE_BPG without inserts. We also successfully applied

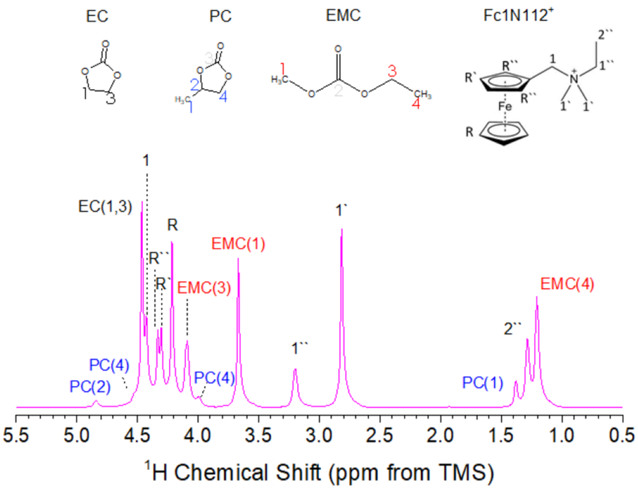

Figure 8. ^{1}H NMR spectrum of 0.85 M [Fc1N122][TFSI] dissolved in EC/PC/EMC obtained using a diffusion probe with a maximum gradient strength of ~31 T/m.[24] The resolution of the spectrum is not comparable with what would be obtained using an ordinary liquid NMR probe due to the magnetic field inhomogeneity induced by the gradient coil, which is not completely ameliorated by the shimming of the magnet. Reprinted with permission from K. S. Han et al. *J. Chem. Phys.* **2014**, *141*, 104509 (ref 24). Copyright 2014 AIP Publishing.

this configuration to various other systems, notably a 20% volume of ethanol mixed with isooctane (viscosity of pure isooctane is 0.473 mPa·s at 25 °C[33]) containing 1% volume of tetramethylsilane (TMS) (Figure 7f).

Artifacts due to the Background Gradient (g_0) and Methods for Suppression. While it has been commonly used for bulk liquid electrolytes for some time, PFG-NMR is finding increasing application among battery scientists across an expanding range of different sample compositions and phases as a technique for diffusion measurements. In particular, this unique technique is occasionally used for the determination of diffusion coefficients of ions/molecules confined in porous materials, such as electrolytes confined in carbonaceous pores. In this case, the measured diffusion coefficient may not be a genuine reflection of the actual translational motion of the diffusing species due to the interference of the background (internal) gradient spanning between the surface and the pore center of the host material. In the PFG-NMR experiment, the applied magnetic field gradient can be viewed as a sort of "ruler" for determining the translational displacement of ions/molecules along the gradient axis during a given time interval. However, the effective background (internal) gradient (g_i) can interfere with the applied gradient from the probe, resulting in a total gradient differing from what the experimentalist assumes has actually been applied (g_a), with a corresponding systematic error introduced into the measured diffusion coefficients. This aspect can make it more challenging to apply PFG-NMR to (magnetically) heterogeneous samples such as viscous liquids, fluids at fluid/solid interfaces, and ions in solid state materials (such as solid-state lithium ion conductors) due to additional attenuation of echo signal induced by these internal gradients and the problem of adequately accounting for the effective gradient arising from the g_i and g_a cross terms.[7,11,34−37] In addition, both the T_1 and the T_2 relaxation times in these materials can be quite short relative to Δ, which essentially precludes echo signal detection after applying Δ in the PFG-NMR sequence.

For samples with short T_2 relaxation times, it may be possible to use a simple spin−echo (SE) PFG sequence (Figure 1a), which is composed of two radio frequency (rf) and two gradient pulses, with significantly increased gradient power to attain sufficient attenuation. However, this simple PFG sequence may be susceptible to the artifact arising from the internal gradient (g_i).[6] If the simple SE sequence is modified to a stimulated echo sequence composed of mono or bipolar gradients as shown in Figure 1b,c, this artifact can be more effectively suppressed.[3] Data obtained utilizing the STE_BPG sequence revealed that it can suppress the internal gradient artifacts of viscous liquids and liquid/solid interfaces containing a solid phase with weak magnetic susceptibility (χ).[16] On the other hand, however, STE_MPG may not be ideal for the suppression of spatially nonconstant background gradients and can result in a dependence of $D_{app}(\Delta)$ on the external magnetic field strength (Figure 9a,b) and anomalous

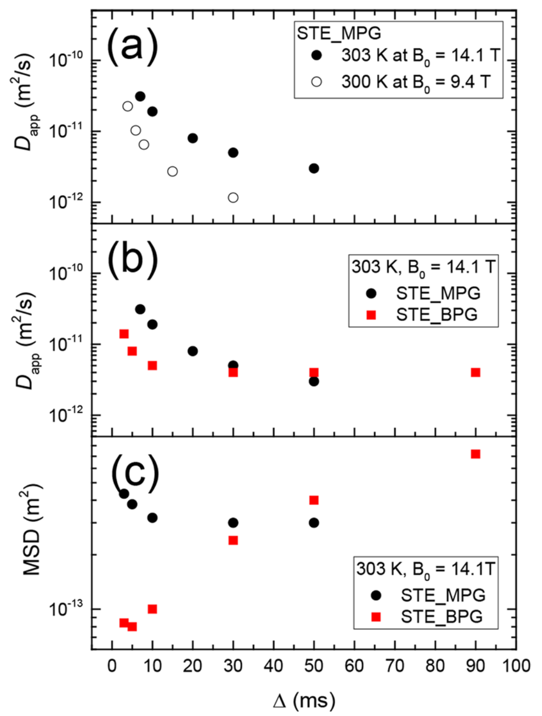

Figure 9. Diffusion time (Δ) dependent (a) apparent diffusion coefficients of $[C_4mim]^+$ cations for $[C_4mim][CF_3SO_2)_2N]$ ionic liquid confined in ordered mesoporous carbon (average pore diameter = 8 nm, surface area = 673 m^2/g, and pore volume = 0.59 cm^3/g) obtained by using STE_MPG under the external magnetic fields of 9.4 and 14.1 T. (b) $D_{app}(\Delta)$ obtained by using STE_MPG and STE_BPG at 303 K under B_0 = 14.1 T. (c) Mean squared displacements of $[C_4mim]^+$ cations calculated from the $D_{app}(\Delta)$ plotted in (b).

behavior in its mean squared displacement (MSD) in the early stage of Δ as shown in Figure 9c. The STE_BPG sequences employing bipolar gradients with symmetric and asymmetric gradient ratios, $g/-g = |1|$ and $|6.7|$, respectively, were originally developed for the suppression of the internal gradient artifact arising due to the nonconstant background gradient.[7,11] For an ionic liquid sample confined in mesoporous carbon, the $D_{app}(\Delta)$ obtained from STE_MPG and STE_BPG exhibited this difference in behavior at the early stage of Δ, while the measured values of the diffusion coefficients converged to the same value at Δ ≥ 30 ms (Figure 9b). Evidently then, great care should be taken with interpreting $D_{app}(\Delta)$ for the study of the pore structures such as tortuosity (T) and porosity (= surface-to-pore-volume ratio, S/V),[21] that can be defined as follows:

$$\frac{1}{T} \equiv \frac{D_\infty}{D_0} \tag{2}$$

$$D_{app}(\Delta)/D_0 = \left[1 - \frac{4}{9\sqrt{\pi}}\frac{S}{V}\sqrt{D_0\Delta}\right] \tag{3}$$

where D_0 and D_∞ are the diffusion coefficient at $\Delta = 0$ (or the bulk unrestricted diffusion coefficient) and at the steady state diffusion where D values reach a plateau, respectively.

As with convection compensation, though, the 13-interval STE_BPG sequence restricts the minimum Δ that can be used during the experiment due to the higher number of rf and gradient pulses between the encoding and decoding gradient pulses and the necessary spacing delays which must be introduced between them. Therefore, it is usually not feasible to use modified STE sequences for samples with short relaxation times (≤4−5 ms), and the STE_MPG sequence is necessitated.[38,39] Indeed, the background gradient artifact is still latent in some cases for highly magnetically heterogeneous samples, such as porous carbon materials,[40] even when using STE_BPG sequences.

Several alternative methods have been developed for the determination of genuine diffusion coefficients with a persistent background gradient artifact in PFG-NMR measurements.[7,11,34−37] However, these proposed methods are not suitable for all systems. As a final recourse when it is suspected that this internal gradient artifact may be impacting the measurements, it can always be diagnosed by measuring the diffusion coefficient with at least two PFG sequences or utilizing two different external magnetic field strengths.[41] If there is no effective internal gradient artifact, the diffusion coefficient will be the same regardless of the external magnetic field strength[41] and the pulse sequence employed.

Artifacts due to Radiation Damping. As pointed out previously, radiation damping can have a significant impact on PFG-NMR measurements of highly concentrated samples. Under such conditions, analysis will be complicated by the precession of spin magnetization that creates oscillating currents in the receiver coil through induced electromagnetic fields.[42] The oscillating current in the receiver coil produces a magnetic field in a process called radiation damping (RD).[43] The strength of the interaction increases with the quality factor of the radio frequency coil, thereby strongly appearing in cryogenic NMR probes.[44] The interaction between the precessing spins and the NMR spectrometer electronics is also particularly apparent at high magnetic field strengths (>14 T).[42]

RD is most readily identified in relaxation measurements, where radiation damping greatly accelerates relaxation processes.[45,46] However, effects of RD are widespread, in part because RD also complicates pulse width nutation experiments, thereby extending effects related to miscalibrations to experiments dependent on accurate pulse widths.[47] Previous literature[48,49] suggested that the diffusion coefficients in simple spin−echo (Figure 1a) based PFG-NMR experiments are distorted by RD. Specifically, the presence of RD can result in increases in signal intensity prior to attenuation as a function of gradient strength in contrast to the typical Stejskal−Tanner relationship, and the deviation in the signal intensity trend is especially apparent when a narrow integral window or peak height is used to calculate the diffusion coefficient.[48]

Various strategies can be used to minimize RD and the effects of RD on PFG-NMR. Strategies include changes to (1) sample geometry and composition, (2) hardware, and (3) pulse sequences. Adjustment to sample geometry and composition are straightforward methods to reduce RD effects. Dilution of the observed spins can be achieved by using solvents with isotopes that are not measured, such as deuterated solvents when acquiring ^{1}H NMR data. Reducing the number of observed spins without the use of isotopes is also possible by reducing the total sample volume by using coaxial inserts to introduce dead volume. The second group of strategies encompasses hardware optimization. The least demanding adjustment to hardware is to use an NMR instrument at a lower field strength. If other magnets are not available, deoptimization of the probe tuning reduces RD effects, although this results in drastic increases in pulse lengths.[50] Alternatively, advanced hardware modifications that employ active electronic feedback on the tuned sample circuit can also suppress the effects of RD.[51] If changes to the sample and hardware are not readily available as solutions, unique pulse sequences have been developed to reduce the effects of RD. For example, pulse sequences that utilize dephasing magnetic gradients to down select the volume of sample measured,[52] thereby reducing the effects of RD in a similar manner to introducing dead volume, have been demonstrated in conventional 1D NMR spectra with possible extension to PFG-NMR. Alternatively, spin−echo based PFG-NMR measurements have been developed in which the Q (quality factor of the r.f. coil) switches during the pulse sequence to attenuate radiation dampening.[49] Stimulated echo based PFG-NMR measurements incorporating a reduced tip-angle for the first pulse have also been demonstrated to reduce effects of RD.[53] In practice, realization of possible distortions from RD are important for accurate diffusion measurements and use of several strategies to reduce RD effects is warranted when measuring concentrated samples at high magnetic fields.

Diffusion Behaviors under Chemical Exchange. Here, it is noteworthy to introduce chemical exchange effects on the PFG-echo profile, revealing a strategy for determining accurate diffusion coefficients for a spin system experiencing chemical exchange between two states, A and B. The PFG-echo profile varies according to the diffusivities (D_A and D_B), populations (P_A and P_B), and mean lifetimes (τ_A and τ_B) of each state. Cabrita et al.[54] showed using simulations that the PFG-echo attenuates almost monoexponentially if the diffusion ratio between the two states is smaller than 5, regardless of the population differences. It also predicted that the diffusion coefficient depends on the relation of diffusion time (Δ) and mean lifetime (τ). In the regime where $\Delta/\tau < 0.1$, where exchange is effectively not present, there will be two individual diffusion coefficients D_A and D_B, while in the limit of $\Delta/\tau > 10$ (fast exchange), the observed diffusion coefficient will be $D_{obs} = P_A D_A + P_B D_B$. For the determination of the correct diffusion coefficients of each state, A and B, the observed PFG-echo signal, M_{obs}, should be deconvoluted to the contribution of the individual states, M_A and M_B, instead of using a simple biexponential relationship, $P_A \exp(-D_A b) + P_B \exp(-D_B b)$, where $b = (\gamma\delta g)^2(\Delta - \delta/3)$.[54]

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Salama et al.[55] observed two pairs of ^{1}H resonances from DME molecules in $Mg(TFSI)_2$/DME solutions due to the slow (relative to the NMR time scale) exchanges between free DME molecules and those coordinating to Mg^{2+} ions. This is a good example to show the difference in diffusivities fitted with and without consideration of exchange between the two different environments. As shown in Figure 10a, we tested diffusion of DME molecules in $Mg(TFSI)_2$/DME solution and showed that, when using an isotropic diffusion model (Stejskal−Tanner equation), the apparent diffusion coefficient $D_{\text{free,DME}}$ decreases with Δ while $D_{\text{bound,DME}}$ increases with Δ until $D_{\text{free,DME}} \approx D_{\text{bound,DME}}$ when Δ becomes considerably longer than the residence times of bound DME (83 ms) and free DME (200 ms). When Δ is comparable to the lifetime of either site, Figure 10b shows the experimental signal attenuation curves at Δ = 80 ms fitting with the Stejskal−Tanner equation (dashed lines) and under conditions with slow exchange (solid lines). It is clear that the fitting to the Stejskal−Tanner equation does not work well when $M_Z/M_0 <$ 0.1 due to the exchange. In this case, an analytical formula for diffusion under slow exchange[56] is necessary to obtain the correct diffusion coefficients of free DME and bound DME at $\Delta \sim 0$.

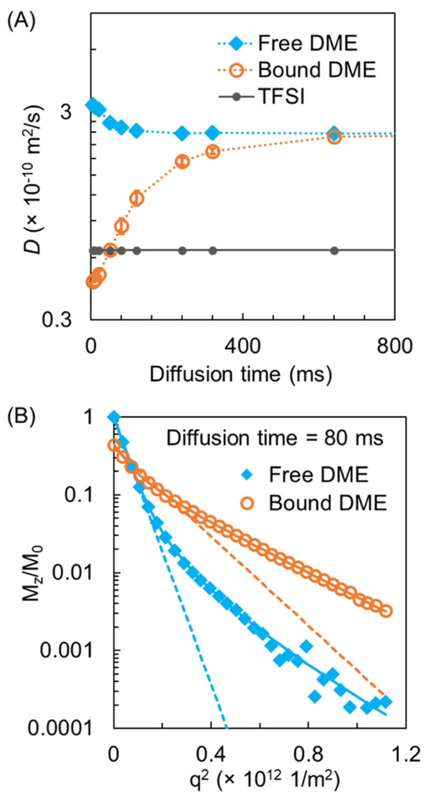

Figure 10. (a) Apparent diffusion coefficients of free DME, bound DME, and TFSI obtained from best fits using Stejskal−Tanner eq 1 for 1.06 m $Mg(TFSI)_2$/DME plotted against diffusion time at varying temperatures −5 °C. (b) ^{1}H PFG NMR signal attenuation curves as a function of "diffusion area" q^2 (= $\gamma^2 g^2 \delta^2$) of free DME and bound DME of 1.06 m $Mg(TFSI)_2$/DME at −5 °C with a diffusion time of 80 ms. Dashed and solid lines are the best fittings to eq 1 and eqs 4 and 5 (diffusion under the conditions of two-site exchange), respectively.

The signal intensities during diffusion measurements under the conditions of slow exchange between the two sites (A and B) can be written as a function of diffusion area ($q^2 = \gamma^2 g^2 \delta^2$), diffusion time (Δ), exchange rates (k_A and k_B), diffusion coefficients of A and B before the exchange occurs (D_A and D_B), and magnetizations of A and B at equilibrium (M_A^0 and M_B^0):[54,56,57]

$$M_A = \left[\frac{M_A^0}{2} - \left(\frac{\delta M_A^0 - k_B M_B^0}{2\lambda}\right)\right] \exp[(-\sigma + \Lambda)\Delta] + \left[\frac{M_A^0}{2} + \left(\frac{\delta M_A^0 - k_B M_B^0}{2\lambda}\right)\right] \exp[(-\sigma - \Lambda)\Delta] \quad (4)$$

$$M_B = \left[\frac{M_B^0}{2} + \left(\frac{\delta M_B^0 + k_A M_A^0}{2\lambda}\right)\right] \exp[(-\sigma + \Lambda)\Delta] + \left[\frac{M_B^0}{2} - \left(\frac{\delta M_B^0 + k_A M_A^0}{2\lambda}\right)\right] \exp[(-\sigma - \Lambda)\Delta] \quad (5)$$

$$\sigma = \frac{1}{2}(k_A + k_B + D_A q^2 + D_B q^2) \quad (6)$$

$$\lambda = \frac{1}{2}(k_A - k_B + D_A q^2 - D_B q^2) \quad (7)$$

$$\Lambda = \sqrt{\lambda^2 + k_A k_B} \quad (8)$$

Here, bound DME coordinates to Mg^{2+} before the exchange occurs, and it is reasonable to assume that $D_{Mg} \approx D_{\text{bound,DME}}(0)$; then, we can estimate D_{Mg} in an electrolyte, indirectly from $D_{\text{bound,DME}}(0)$, which is the apparent diffusion coefficient of DME molecules binding to Mg^{2+} ions at diffusion time Δ = 0 (see Figure 10a).

OTHER NMR METHODS FOR DIFFUSION MEASUREMENT

PFG-NMR Combined with Magic Angle Spinning (MAS) NMR. Combining PFG-NMR with magic angle spinning (MAS) NMR for diffusion measurements in solid-state and gel-like systems provides some advantages relative to static experiments,[58,59] such as gaining NMR sensitivity and peak resolution by reducing line broadening, as well as mitigating radiation damping artifacts.[60] It is still challenging to apply PFG-MAS NMR to diffusion measurements for solid samples due to the weak gradient strength that the specialized probes required can typically supply (up to ∼50 G/cm in an ordinary PFG-MAS probe) coupled with the intrinsically slow diffusion of ions/molecules in typical solid-state samples. For example, $D_{Li} \approx 10^{-18}$−10^{-12} m^2/s within the temperature range of 200−480 K in typical oxide solid-state lithium ionic conductors such as garnet materials.[61] However, PFG-MAS NMR is still quite valuable to utilize for the study of anisotropic diffusion of multiple species (components) confined in porous materials. There are a few examples of PFG-MAS NMR applied to polymer electrolytes[62] and membranes,[58] demonstrating its ability to unravel heterogeneous diffusion behavior of confined solvent molecules[58] and electrolyte components within polymer structures.[62] Notably, it demonstrated that MAS can influence the diffusion coefficient[63] and the structure of the soft polymer[62] due to the centrifugal force generated by fast sample spinning. As a result, PFG-MAS NMR results should be carefully interpreted

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with a consideration of the artifacts that could arise unintentionally due to the presence of centrifugal force.

Relaxation Measurements. As pointed out previously, it is typically quite difficult to apply PFG-NMR to solid state materials, such as solid polymers and solid-state ion conductors, due to the fast T_2 relaxation that even the active, mobile ions in these materials possess. Often other NMR parameters, such as relaxation times (T_1, T_2, and $T_{1\rho}$) and line widths,[64] are therefore used to determine the rotational correlation time (τ_c) of observed spins for the determination of the diffusion coefficient using the Einstein−Smoluchowski equation:

$$D = \frac{a^2}{n\tau_c} \tag{9}$$

where, D, a, and τ_c are the diffusion coefficient, the size of the active ion, and the rotational correlation time, and the coefficient n is 2, 4, and 6 for 1-, 2-, and 3-dimensional diffusion, respectively.[65−67] Since eq 9 strictly applies only in the uncorrelated (i.e., dilute) limit, it is also customary to introduce a multiplicative correlation factor, f (with $0 < f < 1$), to account for the retarding influence of ion−ion correlations, particularly when the ion hopping is defect-mediated (e.g., by vacancies).[68] In some cases, a and τ_c^{-1} in eq 9 can be replaced by the jump distance (l) and the exchange rate (τ_{exch}^{-1}) of ions/molecules hopping between the two sites separated with a distance of l with a time constant of τ_{exch}, which can be determined from 2-dimensional exchange spectroscopy (2D EXSY).[69] Here, it is also worthwhile to introduce the use of the T_2/T_1 ratio instead of time-consuming temperature-dependent T_1, T_2, or $T_{1\rho}$ measurements for the determination of τ_c. If it is possible to assume that both T_1 and T_2 relaxations are mainly correlated with the motion of the observed nucleus, the rotational correlation time can be directly extracted from the T_2/T_1 ratio.[70−72] Using this method, it is still possible to determine the diffusion coefficients of cations and counter-anions in solid polymers. Examples include diffusivity measurements of Li^+ ions from an ionic liquid dissolved into a solid poly(ionic liquid) pentablock terpolymer for use as lithium ion battery electrolytes[66] and Na^+ ions in PEO which had been cross-linked to improve both mechanical robustness and sodium ion conductivity.[67]

Here, we want simply to comment about the T_1 relaxation behavior of quadrupolar nuclei with spin numbers $I = 3/2$ (such as ^{7}Li, ^{23}Na, and ^{131}Xe) and $I = 5/2$ (such as ^{25} Mg and ^{27}Al). On deriving the relaxation rate equations from the quadrupolar Hamiltonian, it becomes clear that the relaxation curves for these quadrupolar nuclei are not generally a single exponential decay, in which case T_1 can be extracted by fitting the relaxation curve using the equations below:[73−75]

For $I = 3/2$,[73]

$$1 - \frac{M(t)}{M(\infty)} = A\exp(-2W_2t) + B\exp(-2W_1t) \tag{10}$$

where $M(t)$ is the NMR signal after time period of t and W_1 and W_2 are the transition probabilities between the Zeeman energy states ($W_1 = 3/2 \leftrightarrow 1/2$ and $-1/2 \leftrightarrow -3/2$ and $W_2 = 3/2 \leftrightarrow -1/2$ and $1/2 \leftrightarrow -3/2$), with T_1 given by

$$T_1^{-1} = \frac{2}{5}(W_1 + 2W_2) \tag{11}$$

Similarly, for $I = 5/2$,[74]

$$1 - \frac{M(t)}{M(\infty)} = 0.03\exp(-t/T_1) + 0.18\exp(-6t/T_1) + 0.79\exp(-15t/T_1) \tag{12}$$

In addition, T_2 relaxation can also have multiple exponential decays when the mobility of ions/molecules is very slow, outside of the extreme narrowing regime.[76,77] The details, which are beyond the scope of this review, can be found in the literature.[76,77] Therefore, if the observed T_1 and T_2 relaxation curves are not a single exponential decay, a high degree of caution should be taken with interpreting the diffusion and T_1 (or T_2) data, since it is difficult to discern if it is due to the presence of heterogeneous environments (i.e., multiple T_1 (or T_2) components with single exponential decays) or due to the intrinsic behavior resulting from the loss of mobilities (i.e., single value of T_1 (or T_2) with multiexponential decays).

Overhauser Dynamic Nuclear Polarization (ODNP) Relaxometry. Dynamic nuclear polarization (DNP) is a technique for enhancing NMR signals by transferring electron spin polarization to NMR-active nuclei using radicals (exogenous or endogenous) distributed around the nuclei.[78,79] Then the DNP-NMR signal intensity depends strongly on the distance between the observed nucleus and the radicals, and this distance-dependent signal intensity in DNP-NMR can be used for diffusion measurements. Here, we introduce and briefly discuss Overhauser dynamic nuclear polarization (ODNP) relaxometry for diffusion measurements. As Saun et al.[80] point out, ODNP-NMR can measure the dynamics of water/proton systems with time and length scales of 0.01−1 ns and ≤1 nm, respectively, in contrast to PFG-NMR which typically probes ranges covering 0.001−1 s and ∼1 μm. As an example, these shorter length and faster time scales in ODNP-NMR made it possible to discriminate the 4 orders of magnitude faster diffusion of water/protons in the vicinity of the sulfonyl ($-SO_3^-$) acid groups and the fluorocarbon surfaces compared with water/proton diffusion through the core of water-channels in Nafion membranes.[81] For this work, Song et al.[81] added two kinds of radicals, those that can block sulfonyl acid groups as well as move freely in the core of water-channels, and assumed that the diffusivities of both radical ions are negligible compared with the diffusivity of water/protons. While ODNP relaxometry requires an NMR spectrometer equipped with DNP capability, as well as the necessity to add radical ions into the sample if the sample does not have the radicals, this method appears to be very promising for studies of ion dynamics over nanosecond and nanometer scales as well as the fine structure of polymers[82] within the polymer electrolytes being either currently used or under consideration for a variety of battery systems.

■ DIFFUSION COEFFICIENT ANALYSIS

In this section, we will turn to discussion of how to extract meaningful information for the system from the measured diffusion coefficient within the limit of the Stokes−Einstein equation, which defines the relationship between the diffusion coefficient of a certain molecule or ion with the hydrodynamic radius r_H in a solution with viscosity (η):

$$D = \frac{k_BT}{6\pi\eta r_H} \tag{13}$$

where k_B is the Boltzmann constant and T is the absolute temperature.[8] While the absolute diffusivities are closely

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related to the viscosity of the solution, the measured diffusivity also depends upon the interactions between the constituent species.[83] In most cases, it is possible to measure the diffusion coefficients of individual components, such as cations, anions, and solvent molecules in an electrolyte solution by using PFG-NMR. Then, the ion−ion and ion−solvent interactions can be roughly estimated from the variations of the diffusion coefficients, which are inversely proportional to the hydrodynamic radii[84,85] or to the effective viscosity (η^*)[24] of the diffusing component. Especially for electrolyte solutions composed of cations whose NMR-active isotopes are so-called quadrupolar nuclei (such as Mg^{2+}, Ca^{2+}, and Zn^{2+}), the diffusion ratio between anions and solvent molecules, $D_{\text{anion}}/D_{\text{solvent}}$ ($\propto r_{\text{H,solvent}}/r_{\text{H,anion}}$), is useful to estimate the degree of cation−anion dissociation and ion solvation[86,87] if the cation diffusion coefficient (D_{cation}) is infeasible to measure with PFG-NMR, as discussed above. The degree of cation−anion dissociation can be more precisely estimated from the comparison between the ionic conductivity measured using impedance spectroscopy (σ_{mea}) and that calculated from diffusion coefficients (σ_{D}) using the Nernst−Einstein formula (eq 14):

$$\sigma_D\ (\text{S/cm}) = \frac{Ne^2}{k_B T}(D_{\text{cation}} + D_{\text{anion}}) \tag{14}$$

The number of ions in the unit volume of V_m (N) can in principle be obtained as $(N_A \cdot x)/V_m$, where x is the ion concentration, N_A is Avogadro's number, and V_m is the molar volume of ions. However, $\sigma_D \geq \sigma_{\text{mea}}$ in all cases due to the naïve assumption of complete (100%) ion dissociation in the calculation of σ_D. Therefore, the ratio of $\sigma_{\text{mea}}/\sigma_D$ ($\equiv \alpha \leq 1$) is the degree of cation−anion dissociation, where $\alpha = 1$ would indicate complete ion dissociation.[23,84,85] Also, it is possible to make some degree of estimate of the contribution of cations to the total ionic conductivity from the transport number of active cations (t_{cation}), which can be calculated as

$$t_{\text{cation}} = \frac{n_{\text{cation}} D_{\text{cation}}}{n_{\text{cation}} D_{\text{cation}} + n_{\text{anion}} D_{\text{anion}}} \tag{15}$$

where n_{cation} and n_{anion} are the number densities (molar ratios) of cations and anions, respectively, in the solution. Care must be taken, though, in connecting PFG-derived diffusivities with quantities in theoretical models of concentrated solutions, such as the Concentrated Solution Theory (CST) approach based on the Onsager−Stefan−Maxwell (OSM) formalism.[88] One method that has shown promise is the use of so-called "effective diffusivities" which combine PFG-NMR diffusion measurements of the anions and cations with conductivity measurements to account for the residual ion pairing fraction,[89−91] and recent evidence from PFG-NMR measurements of lithium-ion battery electrolytes containing lithium bis(trifluorosulfonyl)imide (LiTFSI) and lithium bis-(fluorosulfonyl)imide (LiFSI) salts has demonstrated that these effective diffusivities can be directly linked with the binary OSM diffusivities of CST.[92] The ion−solvent interactions can also be estimated using the effective viscosity (η^*), which can be calculated as follows:[24]

$$D_x = \frac{k_B T}{6\pi \eta_x r_H^x} \quad \text{and} \quad D_{x,\text{solvent}} = \frac{k_B T}{6\pi \eta^*_{x,\text{solvent}} r_H^{x,\text{solvent}}} \tag{16}$$

where $r_H^x = r_H^{x,\text{solvent}}$ and η^* will be

$$\eta^*_{x,\text{solvent}} = \eta_x \frac{D_x}{D_{x,\text{solvent}}} \tag{17}$$

with x = solvent, and D_x and $D_{x,\text{solvent}}$ are the diffusion coefficient of a single solvent and x-component of the mixed solvents, respectively. Then η^* will be varied due to the interaction of the ion with the solvent molecules. The relationship $D \propto 1/\eta$ is also useful to test the behavior of liquid confined in the pores, in terms of determining whether or not liquid-like properties are preserved.[16]

■ APPLICATIONS

Conventional Electrolytes. A key benefit of PFG-NMR is the ability to measure diffusion coefficients for many, if not all, of the components in an electrolyte solution. They can then be used to deduce the physical properties of electrolytes such as the degree of ion dissociation/association, ion solvation environment, and transference number of active ions. In addition to their value in purely physicochemical studies of electrolyte properties to inform design decisions, these measurements could also have considerable utility in parametrizing electrochemical battery models for use in battery management control systems (see Tippmann et al.,[93] as an example of such a parametrization study using electrochemical impedance spectroscopy).

Hayamizu et al.[84,94] correlated ionic conductivity with the degree of ion dissociation (Figure 11) and estimated Li^+ transference numbers (t_{Li}) derived from diffusion coefficients of Li^+ cations, anions, and solvent molecules for 14 carbonate-based electrolytes containing LiTFSI,[94] and similarly for hexafluorophosphate (PF_6^-) anion-based electrolytes: 1 M $LiPF_6$/EC, 1 M $LiPF_6$/PC, 1 M $LiPF_6$/DEC, and 1 M $LiPF_6$/EC+DEC (DEC = diethyl carbonate).[84] From the PF_6^- anion-based electrolytes, they observed that $D_{\text{Li}} < D_{\text{PF6}}$ in the 1 M

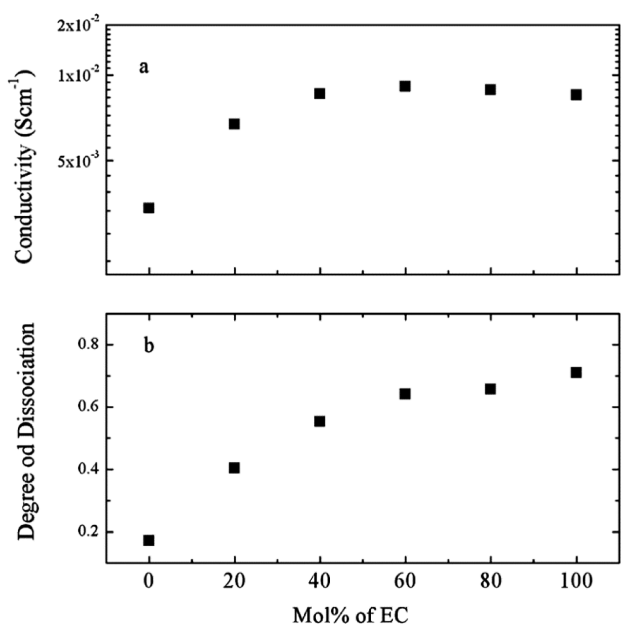

Figure 11. (a) Ionic conductivity and (b) degree of ion dissociation plotted as a function of mol % EC at 303 K for standard lithium-ion battery electrolytes. Reprinted with permission from K. Hayamizu et al. *J. Chem. Eng. Data* **2012**, 57, 2012−2017 (ref 84). Copyright 2012 American Chemical Society.

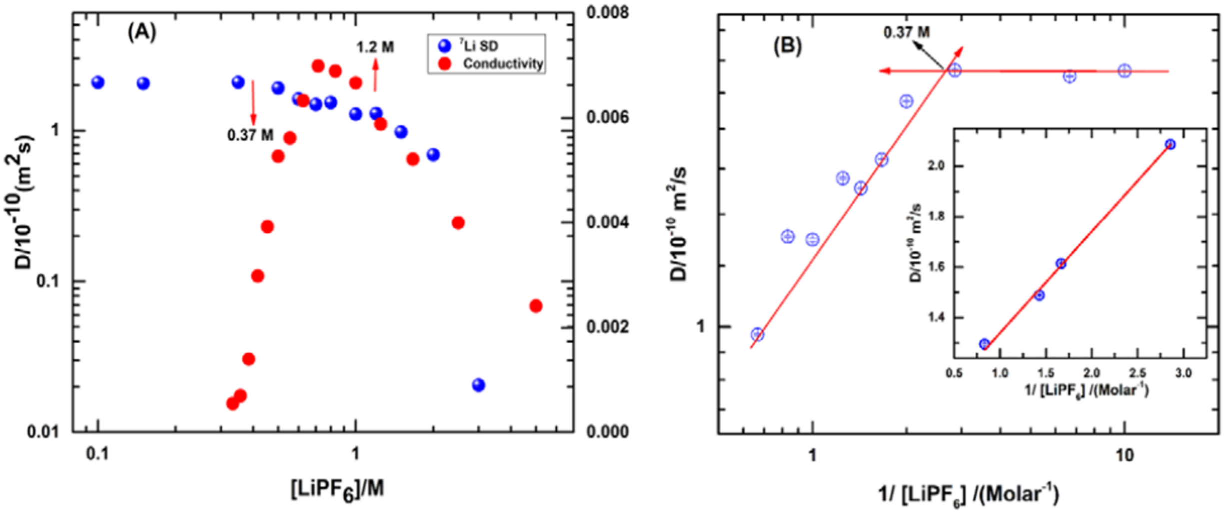

Figure 12. Concentration-dependent ^{7}Li self-diffusion coefficients for $LiPF_6$/PC electrolytes plotted in the ranges of (a) 0.1−3 M and (b) 0.1−1.2 M against the inverse of $LiPF_6$ concentration. The straight lines show the two distinct regimes of ^{7}Li self-diffusion behavior before and after 0.37 M. Reprinted with permission from V. Kumar et al. *J. Phys. Chem. C* **2019**, *123*, 9661−9671 (ref 95). Copyright 2019 American Chemical Society.

$LiPF_6$/EC and 1 M $LiPF_6$/PC solutions whereas $D_{Li} \approx D_{PF6}$ in the 1 M $LiPF_6$/DEC. In the 1 M $LiPF_6$/EC+DEC solutions, D_{Li} gradually decreased with the increase of the EC ratio while D_{PF6} was unchanged. The degree of ion dissociation (α) and ionic conductivity (σ) in the DEC+EC solutions was evidently enhanced by the addition of EC. The ratios $D_{solvent}/D_{Li}$ were almost the same, whereas $D_{solvent}/D_{PF6}$ became smaller with the increase of EC content, suggesting that ion dissociation was boosted due to the improved Li^+ solvation generated by the additional EC. The authors correlated the ionic conductivity and the degree of ion dissociation through this work, as shown in Figure 11. Similarly, Kumar et al.[95] reported the gradual decrease of concentration-dependent ionic conductivity of $LiPF_6$/PC electrolytes as the $LiPF_6$ concentration is increased above ∼1.4 M due to the formation of contact ion pairs (CIPs), correspondingly reducing the proportion of solvent-separated ion pairs (SSIPs) (Figure 12). There are two distinct regimes where the lithium diffusion behavior changes, with the crossover occurring at 0.37 M. This is due to the respective onset of formation of SSIPs, and then subsequently CIPs in the higher-concentration regime. This study also revealed that the Li^+ cation is solvated by approximately four PC molecules, as estimated from the hydrodynamic radius of the solvated Li^+ cation, which was calculated by using the measured D_{Li} in conjunction with the Stokes−Einstein equation. Here, it is important to point out that all ^{7}Li PFG-echo profiles observed in these electrolytes consist of a single exponential decay. This is because, on the time scale for ^{7}Li liquid-state NMR spectroscopy, the interchange of these two different types of lithium ion environments, i.e., SSIPs and CIPs, is sufficiently rapid that they cannot be discriminated, and the resulting signal occurs at the frequency average of the two. As a result, the measured D_{Li} is a population-weighted average of the diffusion coefficients of lithium ions belonging to SSIPs and CIPs, i.e., $D_{Li,SSIP}$ and $D_{Li,CIP}$, as follows:

$$D_{Li} = p_{Li,SSIP}D_{Li,SSIP} + p_{Li,CIP}D_{Li,CIP} \tag{18}$$

where $p_{Li,SSIP}$ and $p_{Li,CIP}$ are the proportion of Li^+ ions in the SSIPs and CIPs, respectively, and $p_{Li,SSIP} + p_{Li,CIP} = 1$. Equation 18 can be modified in terms of the total Li^+ ion concentration, $c_{Li,t} = c_{Li,SSIP} + c_{Li,CIP}$, such that $p_{Li,SSIP} = c_{Li,SSIP}/c_{Li,t}$ and $p_{Li,CIP} = c_{Li,CIP}/c_{Li,t}$, yielding

$$\begin{aligned} D_{Li} &= D_{Li,SSIP}\left(\frac{c_{Li,SSIP}}{c_{Li,t}}\right) + D_{Li,CIP}\left(\frac{c_{Li,CIP}}{c_{Li,t}}\right) \\ &= D_{Li,CIP} + (D_{Li,SSIP} - D_{Li,CIP})c_{Li,SSIP}/c_{Li,t} \end{aligned} \tag{19}$$

Here, $c_{Li,SSIP}$ is the onset concentration for the formation of SSIP. Then $D_{Li,CIP}$ is the intercept obtained from the linear plot of D_{Li} against $1/c_{Li,t}$, as shown in Figure 12b. In addition, Kumar et al.[95] performed 2-dimensional heteronuclear Overhauser spectroscopy (2D HOESY), which is sensitive to the spatially dependent dipolar interactions, between $^{7}Li-^{1}H$ and $^{19}F-^{1}H$, in order to deduce the site dependent local interactions and motional dynamics in the $LiPF_6$/PC solution. The cross-relaxation times were estimated from the cross-peak intensities as a function of the mixing time. The results showed that (1) the ratios of cross-relaxation rates between ^{7}Li and ^{19}F were 6.6−12.7 times larger than the ratio $\gamma(^{7}Li)/\gamma(^{19}F)$ resulting from the stronger cation−solvent interaction, suggesting that these interactions were 15- to 30-fold stronger than anion−solvent interactions, and (2) the lithium interactions with PC are stronger at the H(4) site rather than at the H(1) and H(2) sites (see Figure 8 for site numbering). While the relative cross-relaxation rates determined from HOESY can be used to deduce the preferential interaction of a cation to a certain site of a solvent molecule, it is not straightforward to give a quantitative interpretation for the internuclear distance due to the complexity of intermolecular dipolar interactions combined with rotational and translational motion of molecules, all of which variously depend on the distance r^{-n}, where n = 1 to 6.[95] Some good examples of HOESY combined with PFG-NMR can be found in Pregosin et al.[8]

Feng et al.[96] performed theoretical work for correlating ionic conductivity with the degree of ion dissociation in $LiPF_6$/EC + DEC solutions (1:1 by weight for the solvents) using Onsager−Stefan−Maxwell concentrated solution theory and the generalized Darken relation, in conjunction with PFG-NMR measurements. They calculated ion conductivity with a

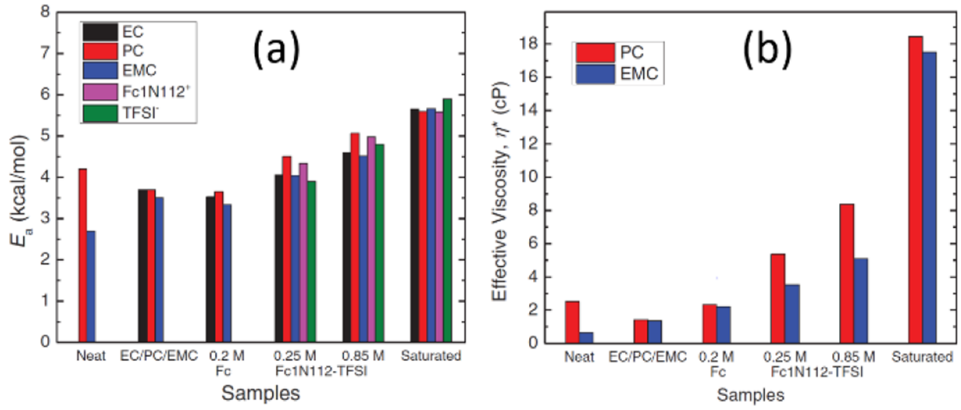

Figure 13. (a) Translational activation energies, E_a, calculated for motion of the solution components for [Fc1N112][TFSI] in EC/PC/EMC from the temperature dependence of the self-diffusion coefficients. (b) Effective viscosities, η^*, for PC and EMC molecules calculated at 25 °C from the measured viscosity, η, and the diffusion coefficients of neat PC and EMC. Reprinted with permission from K. S. Han et al. *J. Chem. Phys.* **2014**, *141*, 104509 (ref 24). Copyright 2014 AIP Publishing.

maximum at 1 M concentration using concentrated solution theory with corrections for the degree of ion dissociation, which ranged from 61% to 37% at $LiPF_6$ concentrations ranging from 0.1 to 1.5 M. Additionally, the lithium-ion transference number was similarly corrected by considering the ion pairing contribution.

Hayamizu et al.[84] also reported that $D_{DEC} \approx D_{EC}$ in a mixture of DEC and EC, while $D_{DEC} > D_{EC}$ (EC melting point is 35−37 °C) as a single solvent, probably due to the presence of the common substructure of the O−C(=O)−O moiety. The same phenomenon was reported by Han et al.,[24] as shown in Figure 13, from the ternary mixture of EC/PC/EMC (4:1:5 by weight), where $D_{EC} \approx D_{PC} \approx D_{EMC}$, while conversely when these are single component solvents $D_{EMC} > D_{PC}$ (D_{EC} is unavailable because it is solid in room temperature). Han et al. used PFG-NMR to determine the diffusion coefficients of cations, anions, and solvent molecules in [Fc1N112][TFSI] dissolved in EC/PC/EMC (4:1:5 by weight) as a function of [Fc1N112][TFSI] concentration at various temperatures. Here, it was found that $D_{PC} < D_{EC} < D_{DMC}$ and the translational activation energy (E_a) and effective viscosity (η^*) of PC became larger than those of EC and EMC after introducing the [Fc1N112][TFSI] into the mixed EC/PC/EMC solvent due to the preferential solvation of $[Fc1N112]^+$ by PC (which was also confirmed with a study of the relaxation properties as well[97]). In the saturated limit for [Fc1N112]-[TFSI], E_a and η^* became similar for all solvent molecules due to the lack of free volume for independent diffusion of the solvent molecules.[98]

Similar preferential solvation of Li^+ cations was observed from electrolyte solutions prepared with binary solvents composed of 1,3-dioxolane (DOL) and dimethoxyethane (DME) (DOL/DME, 1:1 by volume).[85,99,100] In these electrolytes, it was determined that $D_{DOL} > D_{DME}$ after introducing lithium salts, while $D_{DOL} \approx D_{DME}$ in the neat DOL/DME (1:1 by volume) solution. This was also confirmed with ^{17}O NMR spectra, where spectra exhibit a larger shift in the ^{17}O peak of DME while that of DOL remained the same as the single-solvent case. It was therefore assumed that the hydrodynamic radius of DOL ($r_{s,DOL}$) is very close to its actual molecular size (r_{DOL}): $r_{s,DOL} \approx r_{DOL}$ for all samples. Using this fact, Han et al.[85] obtained the viscosity-normalized diffusion coefficients, D_x/D_{DOL}, and the effective hydrodynamic radius, r_x^* (where x = Li, anion, or DME for the electrolytes prepared with various Li salts: lithium bis(trifluorosulfonyl)imide (LiTFSI), lithium bis(fluorosulfonyl)imide (LiFSI), lithium 4,5-dicyano-2-(trifluoromethyl)imidazole (LiTDI), and lithium triflate (LiTfO). It was then possible to correlate the performance of lithuim−sulfur (Li−S) batteries with the normalized diffusion coefficient of the anion ($\propto r_{anion}^*$) measured for a series of electrolytes, as shown in Figure 14.

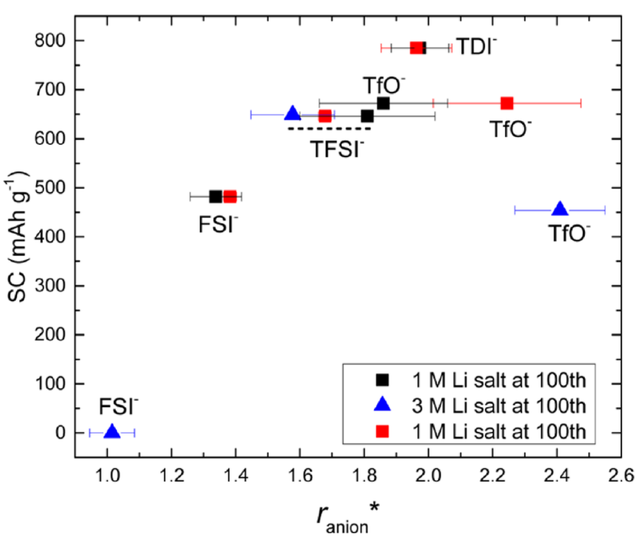

Figure 14. Specific capacity (SC) of Li−S batteries fabricated with electrolytes composed of 1 or 3 M Li salt (LiFSI, LiTfO, LiTFSI, or LiTDI) dissolved in DOL/DME (1:1 by vol) at the 100th cycle plotted as a function of r_{anion}*. The red squares report measurements of SC as a function of r_{anion}* obtained from 1 M Li salt dissolved in DOL/DME with 0.12 M Li_2S_8. Reprinted with permission from K. S. Han et al. *Chem. Mater.* **2017**, *29*, 9023−9029 (ref 85). Copyright 2017 American Chemical Society.

This correlation suggested that certain anions, despite having lower mobility on account of their larger r_{anion}*, could still be promising as an electrolyte for Li−S cells. The TfO^- salt showed worse specific capacity than TDI^- due to its higher activity toward the lithium polysulfides,[99,100] though its mobility is slower compared to TDI^-. The strength of the anion interaction to the Li^+ cation estimated for these anions

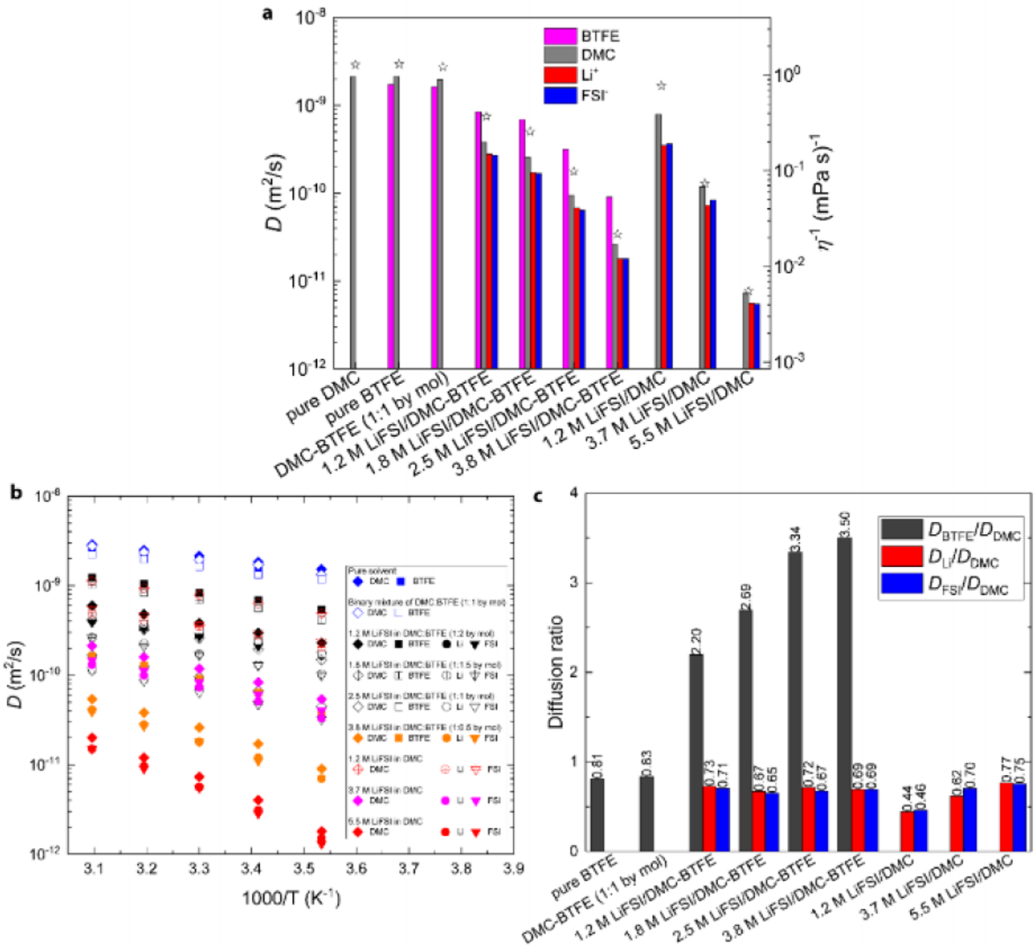

Figure 15. (a) Diffusion coefficients (D) of Li^+, FSI^-, and solvent molecules (DMC and BTFE) at 30 °C plotted along with the inverse of viscosity (η^{-1}) (which is denoted with stars over the corresponding bars), for the LHCE LiFSI/DMC-BTFE. (b) Temperature-dependent diffusion coefficients D_{BTFE}, D_{DMC}, D_{Li}, and D_{FSI}. (c) Diffusion ratios D_{BTFE}/D_{DMC}, D_{Li}/D_{DMC}, and D_{FSI}/D_{DMC} at 30 °C. Reprinted with permission from S. Chen et al. *Adv. Mater.* **2018**, *30*, 1706102 (ref 83). Copyright 2018 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim.

from the ionic conductivity ratio α (= σ_{mea}/σ_D) also supported the existence of a strong TfO^- interaction to Li^+/lithium polysulfides.

Highly Concentrated Electrolytes. The structure and ion dynamics in superconcentrated electrolytes, including localized high concentration electrolytes (LHCE), have also been studied using PFG-NMR diffusion measurements. The concept behind LHCE reported by Chen et al.[83] relied on PFG-NMR to confirm the LHCE solvation structure in the electrolyte solutions, which were composed of lithium bis(fluorosulfonyl)imide dissolved in a mixture of dimethyl carbonate/bis(2,2,2 trifluoroethyl) ether (1:2 by mol)), referred to as LiFSI/DMC-BTFE. The measured diffusivities for these mixtures are compared with the diffusivities of the components and the single-solvent mixtures for a series of salt concentrations in Figure 15, along with the corresponding diffusion ratios.

As predicted by the Stokes–Einstein equation (eq 13), each of the diffusion coefficients of the electrolyte components are globally proportional to the inverse of viscosity (η^{-1}) of the solutions (Figure 15a), while the ratios of the diffusion coefficients between the constituent components are different (Figure 15c) depending on the local ion–ion and ion–solvent interactions. For example, D_{DMC} became smaller than D_{BTFE} after introducing LiFSI salt into the solvent (DMC-BTFE (1:2 by mol)), while D_{DMC} became greater than D_{BTFE}, and the ratio D_{BTFE}/D_{DMC} became larger, as the LiFSI concentration was increased. This strongly suggests that the Li^+ cation solvation occurs mainly by DMC molecules, and BTFE interactions with other components of the electrolyte are very weak. The weak concentration-dependent diffusion ratios D_{Li}/D_{DMC} and D_{FSI}/D_{DMC} suggest that the LHCE solvation structures are mainly composed of Li^+ cations, FSI^- anions, and DMC molecules. They are not sensitive to variations in the population of BTFE in the LiFSI/DMC BTFE electrolytes.

Zeng et al.[101] utilized the highly concentrated electrolyte concept with large salt-to-solvent ratios for a triethylphosphate (TEP) solvent with LiFSI salt in order to enable the design of nonflammable electrolytes for lithium–metal and lithium-ion batteries. PFG-NMR, in conjunction with $^7Li/^{31}P$ NMR and density functional theory (DFT) calculations, was used to develop a molecular-level understanding of the structure of LiFSI/TEP electrolytes over a series of concentrations (Figure 16). It was found that $D_{FSI} > D_{Li} > D_{TEP}$ at lower salt concentration, which ultimately changed to $D_{FSI} \gg D_{Li} = D_{TEP}$ due to complete solvation of Li^+ ions by TEP at concentrations higher than 1.3 (in terms of molar ratios, MR). The $\delta(^{31}P)$ from ^{31}P NMR shift measurements became fixed, whereas the

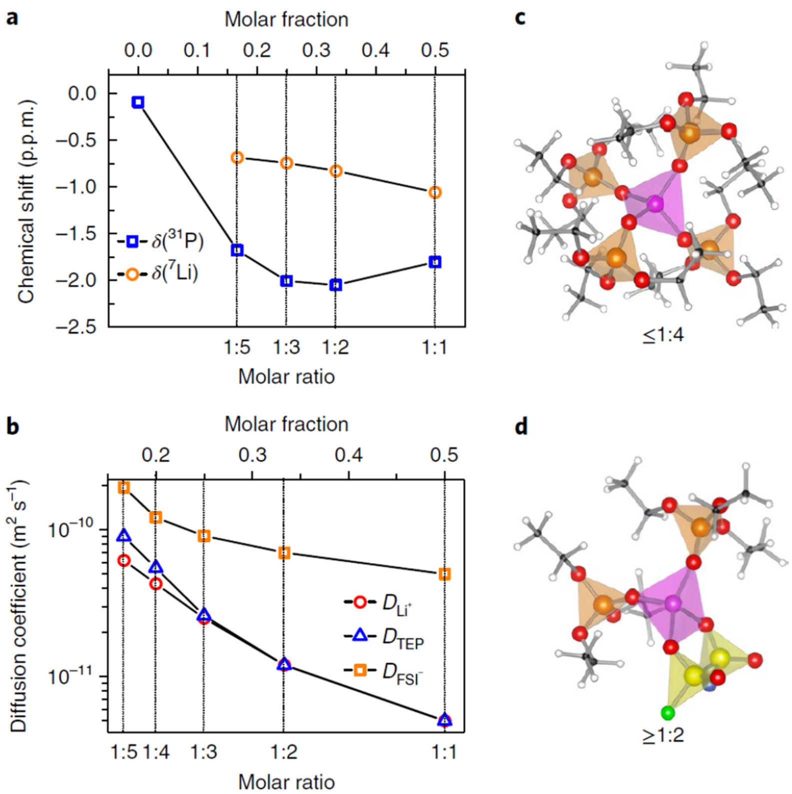

Figure 16. Salt concentration-dependent (a) chemical shifts for ^{7}Li NMR, $\delta(^7\mathrm{Li})$, and ^{31}P NMR, $\delta(^{31}\mathrm{P})$, and (b) diffusion coefficients D_{Li}, D_{FSI}, and D_{TEP} for the high salt-to-solvent ratio electrolyte consisting of LiFSI in TEP. Solvation structure of Li^+ calculated by DFT with (c) four TEP molecules and (d) two TEP and FSI^- in bidentate configuration. The pink, yellow, red, orange, and green spheres represent Li, S, O, P, and F atoms, respectively, whereas ethyl groups are shown using stick models. Reprinted with permission from Z. Zeng et al. *Nat. Energy* **2018**, *3*, 674−681 (ref 101). Copyright 2018 The Author(s).

$\delta(^7\mathrm{Li})$ from ^{7}Li NMR exhibited a gradual negative shift for MR ≥ 1.3. This is due to the increased participation of FSI^- anions in the Li^+ solvation necessitated by the lack of available TEP molecules, consistent with predictions from DFT calculations (Figure 16d).

Li^+ ion dynamics and the structure of "water-in-salt" electrolytes (WiSEs) were studied via analysis of diffusion coefficients obtained from PFG-NMR by Han et al.[102] As shown in Figure 17, the diffusion coefficient decreased with increasing salt concentration, similarly to the previously discussed LHCE. However, for WiSEs it was found that all of the diffusion coefficients were weakly coupled with the inverse of the solution viscosity (η^{-1}), as can be seen in Figure 18. This is due to the phase separation of the mixture into ion-rich and water-rich domains, with the presence of the former helping to facilitate Li^+ diffusion through the latter. This phase separation is also reflected in the increase of the diffusion ratios $D_{\mathrm{Li}}/D_{\mathrm{TFSI}}$ and $D_{\mathrm{H2O}}/D_{\mathrm{TFSI}}$ as the LiTFSI concentration is increased. On the other hand, the $D_{\mathrm{H2O}}/D_{\mathrm{Li}}$ values became smaller due to the increasingly strong interaction of the H_2O molecules with the ion-rich clusters. These two phenomena (phase separation and water interaction to ion-rich clusters) were confirmed by molecular dynamics (MD) simulations, which duplicated exactly the diffusion behavior observed from PFG-NMR as a function of salt concentration. Han et al. also reported that the strong acidity in WiSEs (pH = 2.4 at 20 m (mol/kg) LiTFSI/H_2O) originates from the deprotonation of water molecules in the ion-rich domains by combining the results of PFG-NMR, multinuclear NMR spectroscopy, density functional theory (DFT) calculations, and MD simulations.

It is important to emphasize the importance of PFG-NMR on the study of this kind of electrolyte in collaboration with MD simulations. While the differences in the macroscopic properties predicted by the simulations are scarcely discernible as a function of the simulation parameters, the predicted structures of WiSEs are slightly different at the microscopic scale depending on the force field used for MD simulations.[103−106] The results from PFG-NMR are therefore a key arbiter for validating the reliability of the output from different MD force fields.

Similar phase separation has also been observed from high concentration NaFSI aqueous electrolytes as shown in Figure 19a−d.[107] Further enhancement of the phase separation was observed after introducing $Zn(TFSI)_2$, which had the effect of changing the relative diffusion ratios such that $D_{\mathrm{H2O}}/D_{\mathrm{Na}} < D_{\mathrm{Na}}/D_{\mathrm{FSI}}$ after the addition (with larger $D_{\mathrm{H2O}}/D_{\mathrm{TFSI}}$ also). In 20 m NaFSI without the additive, it was found that $D_{\mathrm{H2O}}/D_{\mathrm{Na}} \approx D_{\mathrm{Na}}/D_{\mathrm{FSI}}$. A decrease of the Zn peak intensity with increasing concentration compared to the actual number of Zn cations in the solution (Figure 19e,f) suggested that about two-thirds of the Zn^{2+} ions become spectroscopically invisible due to the loss of mobility as they are incorporated into a larger network of ions in the 20 m NaFSI + 0.5 m $Zn(TFSI)_2$ mixture. The authors concluded that Zn^{2+} cations and anions confined in the larger networks could "hop" within the network into the electrodes and that fast-moving Na^+ ions in the water channels of the electrolyte networks could protect the Zn

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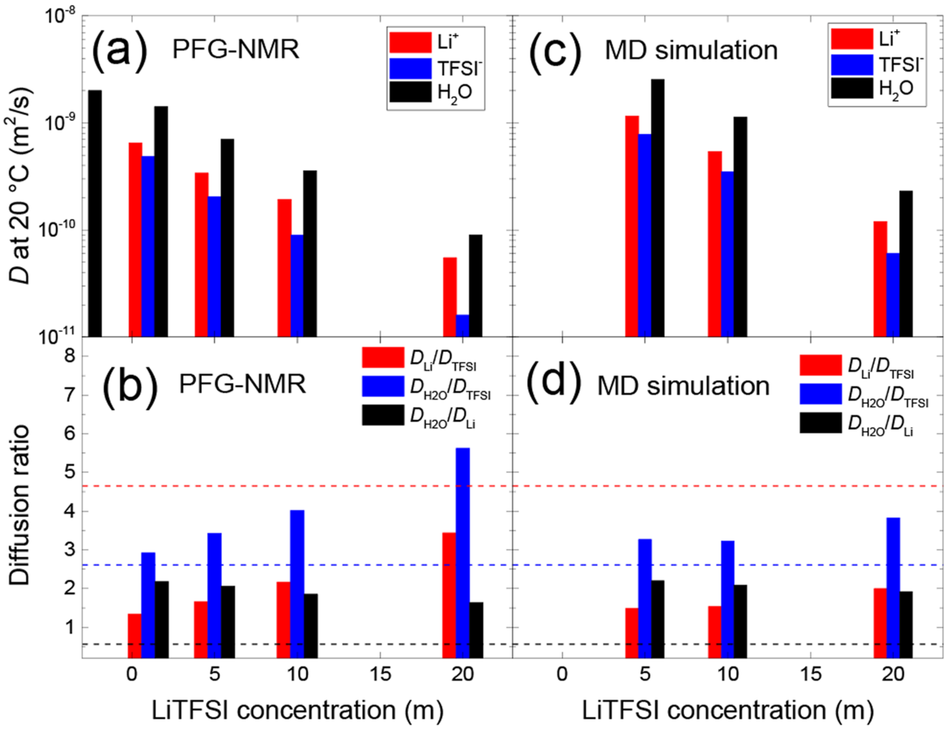

Figure 17. (a) Diffusion coefficients of Li^+, $TFSI^-$, and H_2O and (b) the ratios D_{Li}/D_{TFSI}, D_{H2O}/D_{TFSI}, and D_{Li}/D_{TFSI}, as a function of LiTFSI concentration for a WiSE as measured by PFG-NMR. In part (a), the first black bar shows the diffusion coefficient of pure water ($D_{H2O} \approx 2.0 \times 10^{-9}$ m^2/s) at 20 °C. Parts (c) and (d) report the diffusion coefficients and diffusion ratios, respectively, determined by MD simulations. In part (b) and (d), the horizontal lines show the size ratios r_{TFSI}/r_{Li} (≈ 4.65, red), r_{TFSI}/r_{H2O} (≈ 2.61, blue), and r_{Li}/r_{H2O} (≈ 0.56, black). Reprinted with permission from K. S. Han et al. *J. Phys. Chem. B* **2020**, *124*, 5284−5291 (ref 102). Copyright 2020 American Chemical Society.

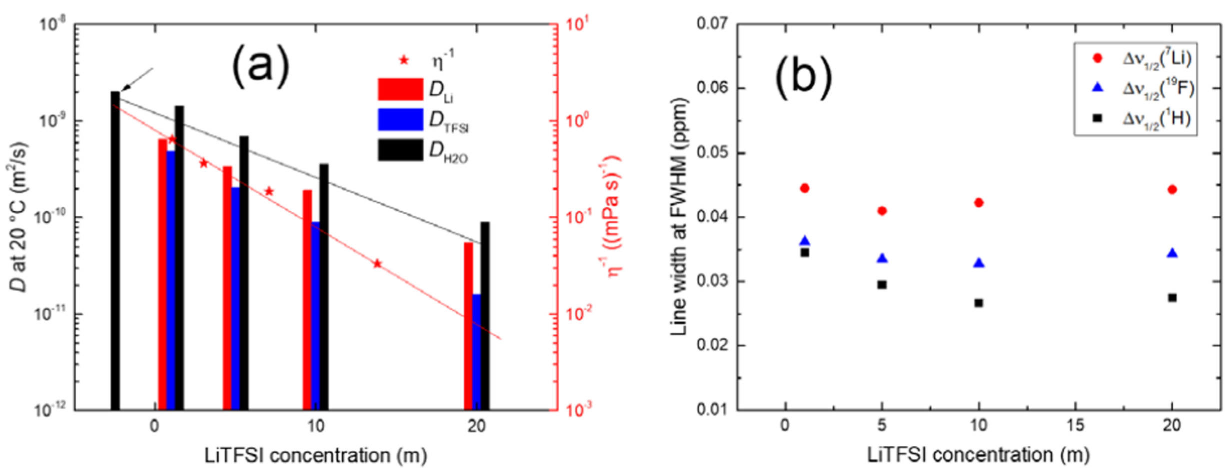

Figure 18. (a) LiTFSI concentration-dependent D_{Li}, D_{TFSI}, D_{H2O}, and η^{-1} obtained from LiTFSI/H_2O solutions. The decrease of diffusion coefficients is slower than that of η^{-1} with the increase of LiTFSI concentration reflecting the minimal effects of viscosity on diffusion behavior of the constituents of LiTFSI/H_2O electrolytes. The diffusion coefficient of pure water at 20 °C is indicated with a black arrow. (b) LiTFSI concentration-dependent line width (full width at half height) of ^7Li, ^{19}F, and ^1H resonances. The narrower line widths at higher concentration also suggest the minimal effects of viscosity on the mobilities of Li^+, $TFSI^-$, and H_2O molecules. Reprinted with permission from K. S. Han et al. *J. Phys. Chem. B* **2020**, *124*, 5284−5291 (ref 102). Copyright 2020 American Chemical Society.

metal anode surface in Zn metal dual-ion batteries employing "water-in-bisalt" electrolytes (WiBSE) composed of 20 m NaFSI and 0.5 m $Zn(TFSI)_2$.

Electrolytes Based on Ionic Liquids (ILs). Room temperature ionic liquids (RTILs) have been considered as an alternative electrolyte system to replace traditional carbonate electrolytes for lithium-ion batteries due to a range of properties such as low volatility, nonflammability, chemical and thermal stability, and wide electrochemical windows which contrast favorably with the conventional electrolyte formulations, combined with high ionic conductivity.[108−111] Physicochemical properties such as these, along with viscosity, density, and diffusivity, can also be significantly altered by modification of the structure of cations and anions. For example, Liao et al.[112] demonstrated that modification of the cation structure by C-2 substitution of the imidazolium cation

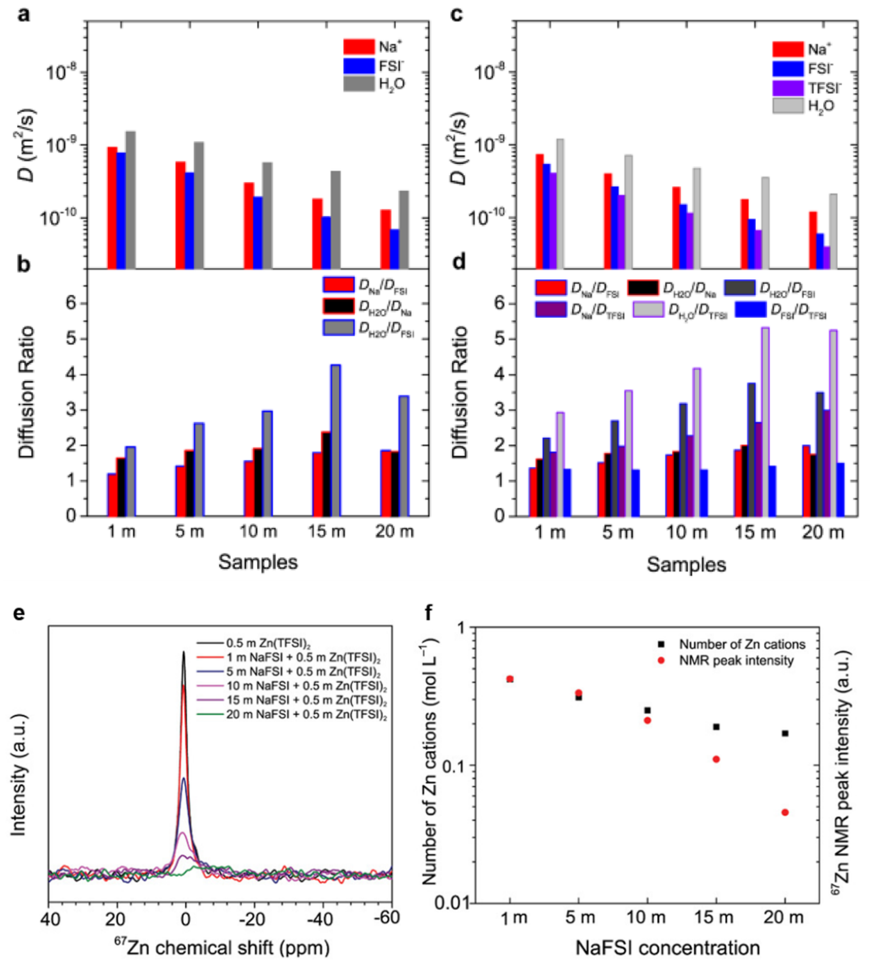

Figure 19. Concentration-dependent diffusion coefficients and diffusion ratios in aqueous NaFSI electrolytes without (a and b) and with (c and d) 0.5 m $Zn(TFSI)_2$, ^{67}Zn NMR spectra (e), and ^{67}Zn peak intensity (integrated area) and nominal concentration (mol/L) of Zn cations (f). Reprinted with permission from I. A. Rodriguez-Pérez et al. *Adv. Energy Mater.* **2020**, *10*, 2001256 (ref 107). Copyright 2020 Battelle Memorial Institute, *Advanced Energy Materials*, published by Wiley-VCH GmbH under the terms of the Creative Commons CC BY license (https://creativecommons.org/licenses/by/4.0/).

in the imidazolium-derived ILs enhances the electrochemical stability by shifting the energy level of the lowest unoccupied molecular orbital. The solvation structure and dynamics of ions dissolved in ILs, as well as the dynamic properties of the ionic liquid itself, have been heavily investigated by PFG-NMR. Tokuda et al. in particular demonstrated the usefulness of PFG-NMR for studying the dynamic and structural properties of ILs using the imidazolium (C_4mim)-based ILs in a series of studies varying the anion species,[113] alkyl chain length in the imidazolium cations,[114] and varying cation species.[115] For the anion series,[113] the summation of diffusion coefficients of the cation and anion decreased by changing anion species according to the following order: $[(CF_3SO_2)_2N]^-$ > $[CF_3CO_2]^-$ > $[CF_3SO_3]^-$ > $[BF_4]^-$ > $[(C_2F_5SO_2)_2N]^-$ > $[PF_6]^-$ (all measured at 30 °C). On the other hand, the ionicity obtained from the ratio of σ_{mea}/σ_D instead decreased as $[PF_6]^-$ > $[BF_4]^-$ > $[(C_2F_5SO_2)_2N]^-$ > $[(CF_3SO_2)_2N]^-$ > $[CF_3SO_3]^-$ > $[CF_3CO_2]^-$. Using the same methods, the alkyl chain length in the imidazolium cation was also investigated. The $C_n$mim notation describes the cation size, where n is the number of carbon atoms in the alkyl chain. Diffusion coefficients observed with a $TFSI^-$ anion for these ILs with the cation series showed that both the cation and the anion diffusion coefficients decreased as per the following order in the cation size: $[C_2mim]^+$ > $[C_1mim]^+$ > $[C_4mim]^+$ > $[C_6mim]^+$ > $[C_8mim]^+$. This trends in an inversely proportional fashion with the viscosity (Figure 20), and the ionicity (proportional to the ratio of molar conductivities, $\Lambda_{imp}/\Lambda_{NMR}$, in this case) also decreased as the chain length was increased.

Binary ILs have also been investigated with PFG-NMR techniques. For a binary IL composed of $[C_2mim][BF_4]$ and $LiBF_4$, measurements of ionic conductivity, viscosity, and ion diffusion coefficients as a function of $LiBF_4$ concentration and at various temperatures were used to understand how the interactions between the IL components influenced the ion conduction mechanism.[116] The results showed that $[C_2mim]^+$ has a different temperature dependence, reflected in its translational activation energy, with increasing $LiBF_4$ concentration compared with Li^+ and BF_4^- (Figure 21). This, in turn, suggested a stronger cation–anion interaction between Li^+ and BF_4^- than between $[C_2mim]^+$ and BF_4^-, and so correspondingly, the ionic conductivity is mainly correlated to the diffusion of $[C_2mim]^+$ cations. Li^+ transference numbers were increased up to ~0.045 at a 1.5 M concentration of $LiBF_4$ from ~0.01 at 0.25 M of $LiBF_4$ as the concentration of $LiBF_4$ was increased. A similar enhancement of the Li^+ cation transference

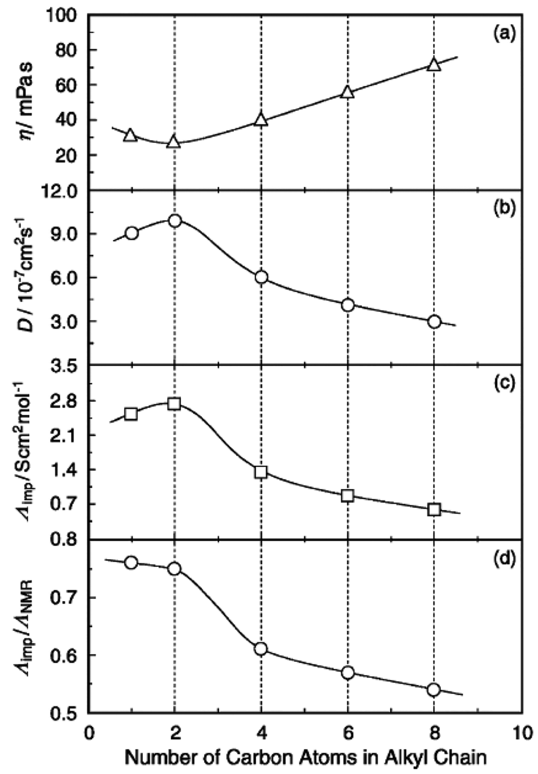

Figure 20. (a) Viscosity, (b) D_{cation} + D_{anion}, (c) molar conductivity (Λ_{imp}), and (d) molar conductivity ratio ($\Lambda_{imp}/\Lambda_{NMR}$) for [$C_n$mim]-[$(CF_3SO_2)_2N$] as a function of alkyl chain length at 30 °C. Reprinted with permission from H. Tokuda et al. *J. Phys. Chem. B* **2005**, *109*, 6103−6110 (ref 114). Copyright 2005 American Chemical Society.

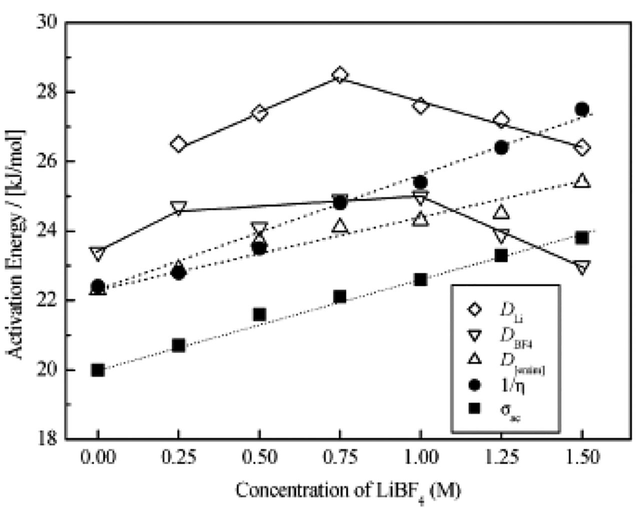

Figure 21. $LiBF_4$ concentration-dependent translational (diffusion) activation energies, viscosity inverses ($1/\eta$), and ionic conductivities (σ_{AC}). Reprinted with permission from K. Hayamizu et al. *J. Phys. Chem. B* **2004**, *108*, 19527−19532 (ref 116). Copyright 2004 American Chemical Society.

numbers (up to 0.132 at 0.337 M of LiTFSI concentration) was also observed from the mixture of *N*-buthyl-*N*-methyl-pyrrolidinium bis(trifluoromethanesulfonyl)imide, [BMP]-[TFSI], and LiTFSI by Frömling et al.[117] They concluded that this is due to the higher number density of lithium ions as compared to other IL electrolytes and then suggested three possible strategies for increasing the lithium ion number density: (i) suppressing crystallization using organic cations and anions with low symmetry; (ii) lowering the cation diffusion relative to that of Li complexes by using bigger organic cations; and (iii) reducing Li complex sizes by using weakly binding anions.

As shown in the previous section, HOESY is also useful to deduce the local structure of the ILs. The local structure and transport properties of the mixture of LiTFSI and *N*-butyl-*N*-methylpyrrolidinium trifluoromethanesulfonimide, [C_4mpyr]-[TFSI], with the molar ratio of 0.1:0.9, were deduced by using HOESY and PFG-NMR, respectively.[118] These experiments showed that the local structure composed of [C_4mpyr]$^+$ and [TFSI]$^-$ was not altered by the presence of Li^+ cations. While the diffusional activation energy of the Li^+ cations is significantly higher due to their much more enhanced diffusivity at higher temperature than that of the [C_4mpyr]$^+$ and [TFSI]$^-$, these latter two have similar diffusion activation energies, suggesting that the Li^+ cation transport mechanism is distinct. In addition, the activation energy calculated from the diffusion coefficient is higher than that calculated from the inverse viscosity (η^{-1}), which suggests the presence of an extra diffusion barrier due to the formation of nanostructures. Based on the above observations, the authors suggested that Li^+ cations hop through the solvation structures by ligand exchange.

NMR Diffraction: Polymer Electrolytes and Liquid Crystals. Polymer electrolytes have received substantial attention as an alternate electrolyte system for lithium-ion batteries, as they can replace volatile, flammable, and toxic conventional carbonate-based electrolytes and can be applied to high energy density all-solid-state batteries (ASSB). These materials consist of a lithium salt intermixed with the polymer matrix and can also produce batteries with increased flexibility, in addition to safety, for novel use cases. Typically, the diffusion coefficients of the molecules confined in the solid pores of a polymer matrix initially decrease and then plateau with increasing diffusion time. This type of Δ-dependent diffusion behavior is so-called "restricted diffusion". This phenomenology is also exhibited by the lithium salts dissolved in polymer electrolytes, and it is observed that the diffusion coefficients of both the lithium ions and the accompanying anions tend to adhere to this restricted diffusion paradigm. For example, diffusion coefficients for the anions in poly(ethylene oxide)-based (PEO-based) electrolytes doped with LiTFSI are shown in Figure 22.[119] The authors compared the diffusion coefficients across the samples at Δ = 70 ms (the bottom of Figure 22), where the plateau starts. From this study, it was revealed that the segmental motions of the polymer chains are strongly correlated with the lithium ion motion in the vicinity of the chains, while anion motion appeared independent from both lithium ion and polymer chain motions. It has also been demonstrated that the structures of the polymer side chains can significantly influence both the cation−anion dissociation and the mobility of the dissociated ions.[119] Timachoca et al.[120] introduced a parameter, *s*, which is the number of Li^+ cations per polymer chain in the mixture of LiTFSI/PEO, and demonstrated that the diffusion coefficients of both the Li^+ cations and the $TFSI^-$ anions decrease with *s* for small values and then ultimately plateau when $s \geq 10$. This finding directly correlated the diffusion of the ions with the polymer molecular weight and the salt concentration.

Similarly, polymer gel electrolytes are also very amenable to study with PFG-NMR methods. Polymer gel electrolytes

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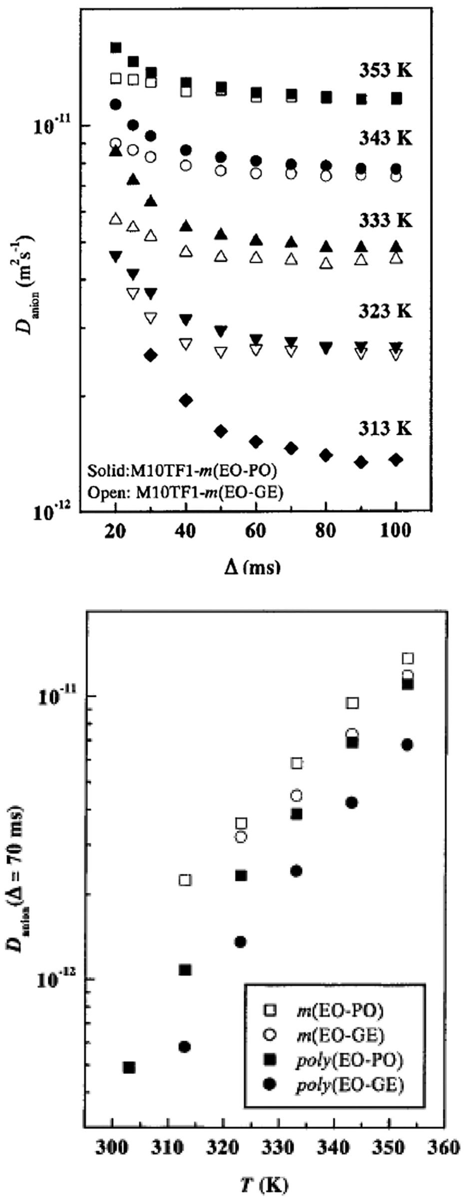

Figure 22. Diffusion delay (Δ) dependent anion diffusion coefficients in a series of poly(ethylene oxide)-based electrolytes doped with LiTFSI (top) and temperature-dependent anion diffusion coefficients at Δ = 70 ms (bottom). Reprinted with permission from K. Hayamizu et al. *J. Phys. Chem. B* **2002**, *106*, 547−554 (ref 119). Copyright 2002 American Chemical Society.

utilize only a small percentage (e.g., by weight) of plasticizer to entrap a liquid electrolyte and produce properties intermediate to both bulk liquid electrolytes and solid polymer electrolytes.[121] In particular, a polymer gel electrolyte composed of poly(ethylene glycol) dimethacrylate and $LiBF_4$−EC/EMC showed the Li^+−BF_4^- dissociation increased and the translational activation energy of the lithium ions decreased as the polymer gel content was increased, which the authors correlated to the segmental motion of polymer chains interacting with lithium cations through the Coulomb interaction and influencing lithium ion hopping between chains.[122] The ionic conductivity and the degree of cation−anion dissociation of the polymer electrolytes can be closely correlated with the number ratio between oxygen atoms in the polymer chains and lithium cations,[122,123] which provides a means of tuning the ionic conductivity and the degree of cation−anion dissociation in polymer electrolytes. This can be accomplished by varying the interaction strength between the constituent components by adding different types of ions as, effectively, cosalts,[124] and by modifying the polymer structures themselves.[119,125] Here we would like to mention that tortuosity and pore size determined from $D_{app}(\Delta)$ (see eqs 2 and 3) may vary with temperature due to the change of $D_{app}(\Delta)$ patterns as a function of temperature as shown in Figure 22.

One important behavior which can be monitored in PFG-NMR measurements for polymer electrolytes in particular is *NMR diffraction*,[22] occurring on account of polymer structure inhibiting the diffusion, such as an aggregation unit of entangled side chains[126] acting as a diffusion barrier. The disruption of the actual diffusion path of the dissolved ions moving through the vicinity of and/or the interior of these polymer structures superimposes an additional oscillatory component on the PFG-echo decay, as shown in Figure 23.[127−129] From this behavior, it is possible to elucidate the

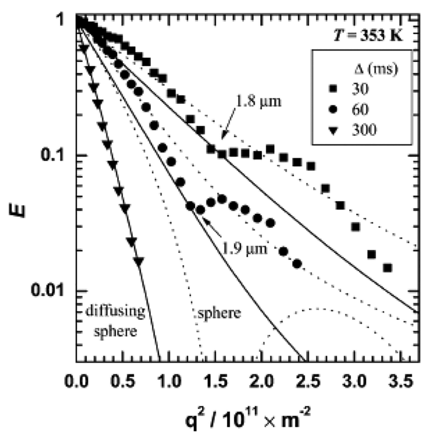

Figure 23. ^{7}Li PFG-NMR echo profiles (E) as a function of Δ obtained from an ethylene oxide-*co*-propylene oxide (m(EO)-(PO), MW ∼ 8000) electrolyte doped with LiTFSI with an Li/O ratio of 0.05, measured at 353 K. Reprinted with permission from K. Hayamizu et al. *Macromolecules* **2003**, *36*, 2785−2792 (ref 128). Copyright 2003 American Chemical Society.

structure of the polymers and the correlated ion dynamics using eq 20,[22]

$$E(q, \Delta) = |s_0(q)|^2 \times \exp\left[-\frac{6D_{\mathrm{eff}}\Delta}{b^2 + 3\xi^2}\left(1 - \exp(-2\pi^2 q^2 \xi^2)\frac{\sin(2\pi qb)}{2\pi qb}\right)\right] \quad (20)$$

where the local structure factor of s_0 is

$$s_0(q) = \frac{\sin(\pi qa)}{\pi qa} \quad (21)$$

In eq 20, ξ is the standard deviation of the mean pore spacing distance of b, a is the pore radius, D_{eff} is the diffusion coefficient for migration between the pores, and Δ is the diffusion time. In most cases, NMR diffraction can detect the influence of structures with size on the order of micrometers, with the sensitivity of the diffraction varying with the PFG-NMR parameters such as diffusion time (Δ),[23,127,130] along

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with the structural heterogeneity.[23,127] In the study of polymer electrolytes, the definition of D_{eff} under the NMR diffraction formalism shows that it is closely related to the diffusion of ions in the vicinity of polymer chains and can therefore be different from the bulk diffusion of ions that belong to the voids.[23] The correlation of ion diffusion with the structural and dynamical properties of polymer chains can be more directly deduced from NMR-diffraction analysis than standard Δ-dependent diffusion behavior. The latter is also much more susceptible to the influence of experimental conditions such as temperature, as discussed previously (Figure 22a). Also, NMR-diffraction analysis can give further information for the size of voids and polymer clusters that cannot be deduced by conventional, variable-Δ PFG-NMR analysis.

The results for various types of polymer electrolyte[127−129] have demonstrated that the doped ions diffuse through the voids between polymer domains (which have a size on the order of a few micrometers) and that the diffusional motion of Li^+ cations is closely correlated with the chains, while anion diffusion is relatively free from interactions with the Li^+ cations and the chains.

Park et al.[23] analyzed ion−ion interactions and resulting structure formation in the mixtures of LiTFSI salt and 1,3-dimethylimidazolium 2-methoxy(2-ethoxy(2-ethoxy(2-ethoxy)))ethylphosphite, [DMIm][MPEGP], ionic liquid as a function of LiTFSI concentration. The results demonstrated several features dictating the overall ion transport in these electrolytes. First, the Coulomb interaction between Li^+ cations and $[MPEGP]^-$ anions is the origin of both the solvation of Li^+ cations in the lower LiTFSI concentrations ($x < 0.4$) and the formation of smectic ionic liquid crystals with a specific stoichiometry, Li^+:$[MPFGP]^-$:$TFSI^-$ = 3:4:1 up to the size of ∼2 μm at higher concentrations ($x \geq 0.4$). As a result, the ionic liquid crystal possesses a negative charge, $[Li[MPEGP]_{1.26}TFSI_{0.32}]^{0.58-}$. Second, NMR diffraction was observed from the ^1H ($[DMIm]^+$ cations) and ^{19}F ($TFSI^-$ anions) PFG-echo profiles in the $x \geq 0.4$ regime, but these were weaker than that observed from ^7Li PFG-echo profiles, which suggests a relatively stronger interaction of Li^+ cations to the surface of negatively charged ionic liquid crystals. Finally, the variation of the diffusion ratios between the mobile ions (i.e., those which are not participating in the structure formation) suggests that those ions preserve bulk ionic liquid properties. The interactions between the constituent ions in these preserved regions of bulk ionic liquid order in the void spaces between the microscale structures depend on the composition (i.e., relative number) of ionic species remaining there.

Solid Lithium-Ion Conductors. The recent explosion in the discovery of fast lithium-ion conductors has broadened the applicability of PFG-NMR for diffusion measurements in the solid state, with the comparatively high mobility of the ions tending to produce relaxation times—particularly longer T_2 times—which are amenable to the technique. Early examples include the work of Hayamizu et al., which succeeded in carrying out ^7Li PFG-NMR on a garnet-type $Li_{6.6}La_3Zr_{1.6}Ta_{0.4}O_{12}$[38] and thiophosphate $(Li_2S)_7(P_2S_5)_3$.[131,132] These materials showed an NMR diffraction behavior probably due to the influence of the microscale material structure on the diffusion paths, as subsequent work on diffusivity measurements of single-crystal $Li_{6.5}La_3Zr_{1.5}Ta_{0.5}O_{12}$ has shown no Δ-dependent diffusion over a range of 20−1000 ms but with some deviations in a crushed powder version of the same sample.[133] In addition, some samples exhibited gradient strength-dependent diffusion behavior, as shown in Figure 24, which is similar to the

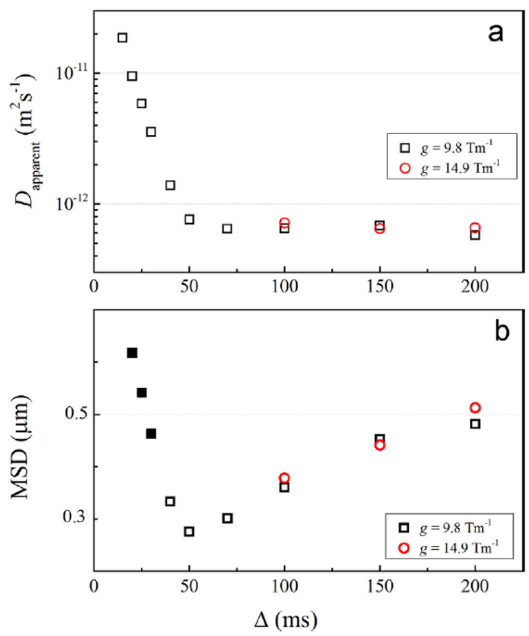

Figure 24. (a) D_{app} and (b) MSD vs Δ at 353 K for Li ions in $LiLa_3Zr_{1.6}Ta_{0.4}O_{12}$[38] at g = 9.8 T/m (square) and 14.9 T/m (circle). The solid squares denote that diffraction patterns were observed. Reprinted with permission from K. Hayamizu et al. *Solid State Nucl. Magn. Reson.* **2015**, *70*, 21−27 (ref 38). Copyright 2015 Elsevier Inc.

diffusion behavior also observed for ionic liquids under the presence of internal gradients, as discussed in the preceding section. Subsequent solid-state PFG-NMR measurements in the thiophosphate ion conductors have aided efforts to both understand the origin of their anomalously high conductivities and further optimize their design. Kuhn et al. measured diffusivities comparable to liquid-state lithium ion electrolytes in the $Li_{10}GeP_2S_{12}$ (LGPS) ion conductor[134] and linked the high uniformity of the dynamics across local and micrometer-size length scales with its excellent conduction. PFG-NMR has continued to find use in transport characterization of related LGPS derivatives such as $Li_{10}SnP_2S_{12}$[135,136] and $Li_{11}Si_2PS_{12}$,[137,138] although in these derivatives a much larger disparity between grain and grain-boundary conduction processes has also been noted. Combined ^7Li PFG-NMR and relaxometry were also recently finally applied to the much more slowly conducting β-Li_3PS_4 (the parent phase for LGPS and its derivatives), yielding a diffusivity of 9×10^{-14} m^2/s, near the limitations of present instrumentation.[139] The ion conductivity estimated using these diffusivity measurements and the Nernst−Einstein equation (eq 14) agreed very well with the directly measured ionic conductivity from electrochemical impedance spectroscopy (EIS).

A related class of thiophosphate solid-state lithium-ion conductors are the argyrodites, Li_6PS_5X, where X = Cl, Br, and I,[30] where in contrast to the tunnel-like intragrain diffusion pathways of the derivatives, the primary structural motif consists of "cage-like" structures with rapid Li ion interchange between vacancy sites and long-range ion conduction mediated by hops between the cages. Recently, Adeli et al.[30] demonstrated that chloride enrichment, $Li_{6-x}PS_{5-x}Cl_{1+x}$, yields a 4-fold increase in the already high ionic conductivity. There are two mechanisms by which the conductivity can be enhanced via this substitution method: reduced interaction between the ions and the anion sublattice as a consequence of the lower polarizability and the introduction of additional vacancies to the cation sublattice. By combining ^{7}Li PFG-NMR and MAS NMR, the authors were able to demonstrate a crossover in the relative importance of these effects, with the isotropic chemical shift trend related to the lattice polarizability more dramatic at $x < 0.25$, whereas at $0.25 < x < 0.5$, calculation of the Haven ratio indicated the increased influence of the vacancy population on the enhanced ion transport (Figure 25). The Haven ratio is calculated as $H_R = D_{PFG}/D_\sigma$,

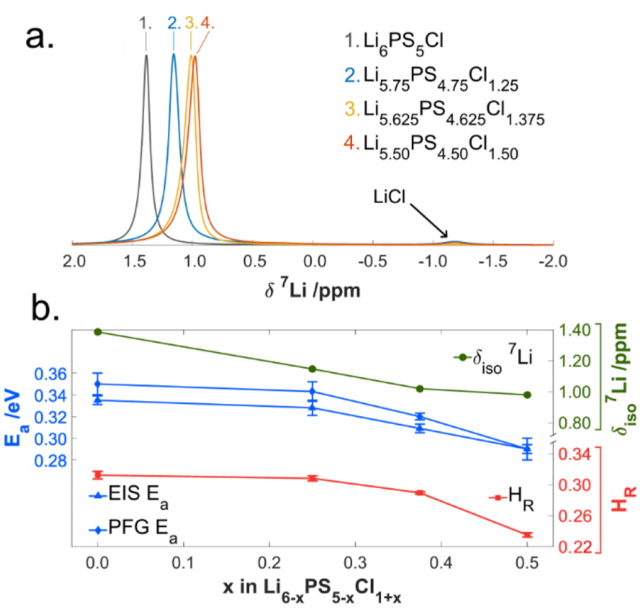

Figure 25. (a) ^{7}Li MAS NMR for $Li_{6-x}PS_{5-x}Cl_{1+x}$ (x = 0, 0.25, 0.375, 0.5), demonstrating the eventual saturation of the isotropic shift trend with increasing x. (b) Comparison of the activation energies derived from ^{7}Li PFG-NMR and EIS, correlated with the isotropic shift trend and the Haven ratios as a function of x, demonstrating the interplay of lattice polarizability and vacancies on the ion transport enhancement of halide-enriched argyrodites. Reprinted with permission from P. Adeli et al. *Angew. Chem. Int. Ed.* **2019**, *58* (26), 8681−8686 (ref 30). Copyright 2019 Wiley-VCH Verlag GmbH & Co.

where D_σ is the diffusivity defined by the ion conductivity obtained from the Nernst−Einstein equation (eq 14), and it provides a measure of correlation in the ion transport, with the increasingly low values of H_R with increasing x signaling that the corresponding enhancement of the ion transport is driven by the increased vacancy population.

These authors also explored the role of aliovalent cation doping in the argyrodite system with PFG-NMR and extended the analysis to examine the influence of mechanical compression on the measured diffusivity values.[140] Using the fast-diffusing (1.21×10^{-11} m^2/s at RT) $Li_{5.35}Ca_{0.1}PS_{4.5}Cl_{1.55}$ composition, variable-temperature ^{7}Li PFG-NMR and relaxometry were performed for both the as-synthesized powder sample and a pellet sample compressed at high pressure (Figure 26). By contrasting the Arrhenius plots for these measurements between the two samples, the authors were able to identify two distinct ion transport regimes: a "high temperature" regime with grain-dominated conduction and a

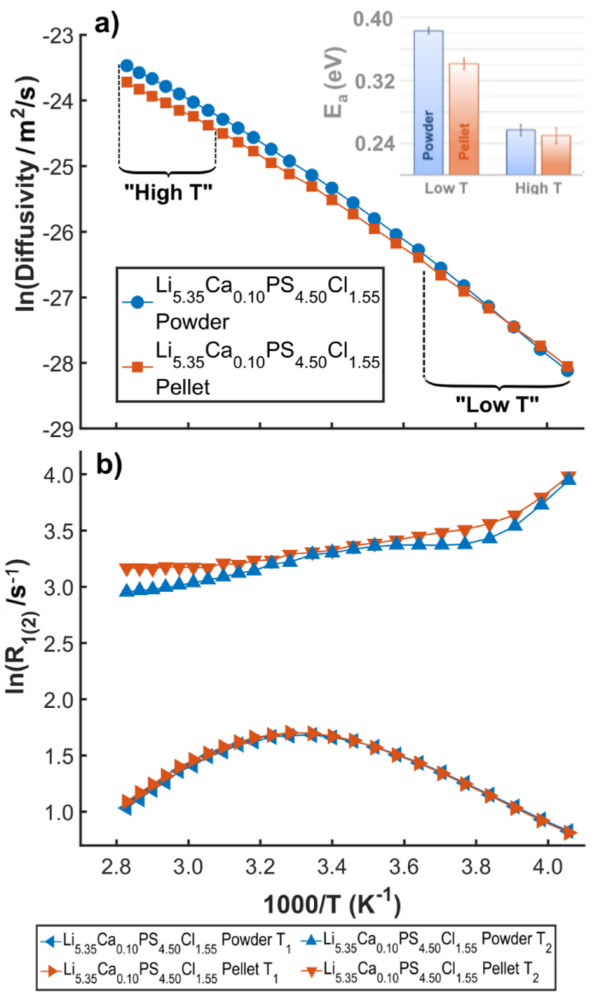

Figure 26. (a) Diffusivity Arrhenius plots from variable-temperature ^{7}Li PFG-NMR measurements on powdered and pressed-pellet versions of $Li_{5.35}Ca_{0.10}PS_{4.50}Cl_{1.55}$, showing nonideal Arrhenius behavior with a crossover in slopes from low to high temperature, which is significantly reduced in the pellet version of the sample. Measurement uncertainties are within the marker size. (b) NMR relaxation rates of the powdered and pelletized $Li_{5.35}Ca_{0.10}PS_{4.50}Cl_{1.55}$, where the presence of two distinct inflection points in the T_2 data supports the PFG-NMR determination that two distinct time scales of motion occur. Reprinted with permission from P. Adeli et al. *Chem. Mater.* **2021**, 33 (1), 146−157 (ref 140). Copyright 2021 American Chemical Society.

"low temperature" regime with grain-boundary-dominated conduction. The activation energies from the PFG-NMR measurements were identical within uncertainty between the two samples in the grain-dominated region, whereas a clear influence of the compression, reducing the grain boundary hopping activation, was observed in the "low temperature" regime. Given that poor interfacial contacts make EIS experiments challenging without considerable applied pressure for these materials, this study demonstrates that PFG-NMR can shed light on micrometer-scale transport processes and mechanochemical interface modification over a range of conditions not accessible to conventional electrochemical techniques.

Here, it is noteworthy to introduce how to determine anisotropic diffusion coefficients, which can be expected to arise in some solid samples. If the diffusion probe has the capability to generate x and y gradients, along with the standard z gradient, it is possible to perform three-dimensional PFG-NMR experiments for diffusion measurements in anisotropic materials without sample rotation.[17,18,141] However, in general, ordinary diffusion probes can only generate the gradient along the z-axis. In such a case, and with powder samples, anisotropic diffusion can be described by the diffusion tensor, which has principal components D_{XX}, D_{YY}, and D_{ZZ}. The tensor components can be obtained from the echo decay as[142143]

$$S(g\delta, \Delta) = \frac{1}{4\pi}\int_0^{2\pi}\int_0^{\pi} \exp[-(\gamma\delta g)^2\Delta(D_{XX}\sin^2\theta\cos^2\varphi + D_{YY}\sin^2\theta\sin^2\varphi + D_{ZZ}\cos^2\varphi)]\sin\theta\, d\theta\, d\varphi \quad (22)$$

for overall orientations (θ, φ) of crystallites in a powder with $0 \le \theta \le \pi$ and $0 \le \varphi \le 2\pi$. If a system has cylindrical symmetry, eq 22 can be reduced to

$$S(g\delta, \Delta) = \frac{1}{2}\int_0^{\pi} \exp[-(\gamma\delta g)^2\Delta(D_{ZZ}\cos^2\theta + D_{XY}\sin^2\theta)]\sin\theta\, d\theta \quad (23)$$

with D_{XY} representing the diffusion component perpendicular to the gradient axis.

In Situ PFG-NMR. Recently, it was demonstrated that both the pore structure of a carbon electrode and ion dynamics in the working electrode can be studied in situ utilizing PFG-NMR.

Forse et al.[144] studied electrolyte dynamics in the supercapacitors fabricated with 0.75 and 1.5 M tetraethylphosphonium tetrafluoroborate (PEt_4BF_4) dissolved in deuterated acetonitrile (CD_3CN) as a function of charge state (Figure 27). Measurements revealed biexponential decays in the PFG-echo profiles, and the authors interpreted this behavior as due to in-pore and ex-pore electrolyte exchange that facilitates the diffusion, resulting in the fast decay in the early stage of b (Figure 27c). This exchange was also confirmed by 2-dimensional exchange spectroscopy (2D EXSY). Based on the diffusion behaviors of cation, anions, and solvents and depending on the voltage of the cell and electrolyte concentration, the authors found a correlation of non-equilibrium charge storage mechanisms with the slower pore diffusion ($D_{\text{in-pore}}$) in the smaller pores. Also, $D_{\text{in-pore}}$ varies due to the ion−ion interactions and ion populations in the pore. Moreover, the authors suggested that because the charging rate of the carbon electrode is proportional to $D_{\text{in-pore}}$, the use of the carbon electrodes with high porosity combined with electrolytes with lower concentrations of ions would accelerate charging processes.

Torayev et al.[145] performed *in situ* ^{1}H PFG-NMR as a function of the discharge depth using the STE_MPG sequence on a Li_2O_2 cell fabricated with a self-standing Super P carbon cathode that discharged with the electrolytes, which consisted of LiTFSI dissolved either in tetraglyme or in dimethoxyethane. From these measurements, it was possible to understand the electrode microstructure, such as porosity and tortuosity. The latter is closely related to the discharge capacity of the Li−O_2 cell, correlated with the accumulation of discharge products in the cathode. The authors analyzed $D_{app}(\Delta)$ and found that the larger tortuosity in the higher discharge depth is due to both pore shrinkage and clogging of the electrode pore by discharge products. The clogging seemed

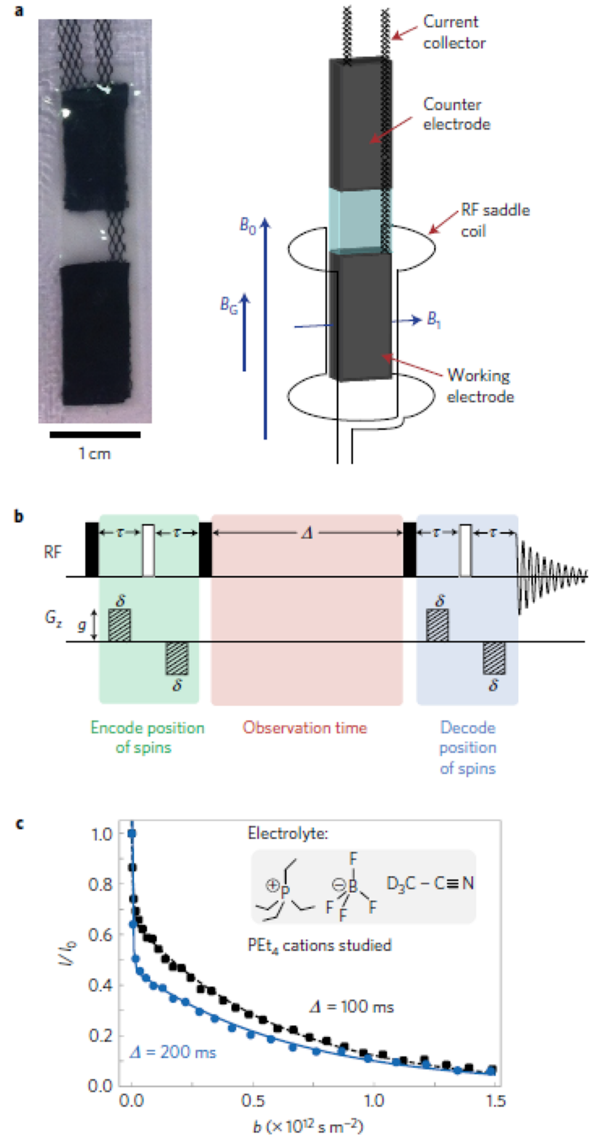

Figure 27. (a) Photograph of a supercapacitor pouch cell and a schematic of the in situ PFG-NMR setup. (b) Pulse sequence; STE_BPG used in this study. (c) Example PFG-echo decays normalized with in-pore cation signal intensity plotted as a function of b.[144] Reprinted with permission from A. C. Forse et al. *Nat. Energy* **2017**, *2*, 16216 (ref 144). Copyright 2017 Nature Publishing Company.

more severe in the electrodes discharged with DME-based electrolyte than in those with tetraglyme-based electrolyte, judged from the relatively higher effective diffusion coefficients from the electrodes discharged with tetraglyme-based electrolyte. This is probably due to the formation of a larger Li_2O_2 particle by DME than by tetraglyme, due to its larger donor number. In addition, the authors claimed that the biexponential decay in the PFG-echo profiles reflect an exchange between the electrolyte clogged in the pore and that existing freely outside of the pore structure over the PFG-NMR time scales. However, the $D_A(\Delta)$ and $D_B(\Delta)$ values obtained from the biexponential fitting were parallel to each other up to $\Delta = 1.2$ s. As a result, in this system the exchange time constant τ will be larger than 12 s based on the Δ/τ-dependent diffusion simulation done by Cabrita et al.,[54] which showed that there are two distinctive diffusion coefficients in the slow exchange limit (where $\Delta/\tau < 0.1$). It is important to emphasize that, in studying confined diffusion in carbon-based materials, the strong magnetic susceptibility[40] can generate significant background gradients, and the exchange of in-pore and out-of-pore ions can also significantly complicate the fitting of PFG echo attenuation curves.

Another new application of in situ PFG-NMR measurements has been the extraction of electrolyte transport properties with spatial resolution in operating electrochemical cells, particularly as applied to lithium-ion chemistries. Following the initial demonstration of magnetic resonance imaging (MRI)—a gradient NMR technique with underlying similarities to PFG-NMR—to capture evolving concentration gradients by Klett et al.,[146] Krachkovskiy et al. (2013) utilized slice-selective PFG-NMR to directly measure the local diffusion coefficient at selected locations along the electrolyte concentration gradient.[147] While diffusion coefficients in these systems can be extracted from numerical analysis of time-resolved images of the concentration gradients[146,148,149] or from inverse modeling approaches,[150,151] these experiments illustrated that the spatial variation—and hence, the concentration dependence—of the diffusion coefficients could be accessed directly. Krachkovskiy et al. (2016) subsequently provided fully realized spatially resolved measurements of both anion and cation diffusion coefficients and transference numbers by combining PFG-NMR and MRI within the same experiment, with a STE_BPG pulse sequence at a particular gradient strength substituting for the initial excitation pulse in a phase-encoding MRI sequence, thereby yielding a pseudo-three-dimensional experiment with chemical shift, diffusion gradient, and imaging gradient dimensions.[152] The authors demonstrated that, despite the considerable concentration gradient formed at the steady state (0.78−1.27 M) when the positive and negative ion fluxes reached a balance, with accompanying inverse gradient in diffusivities, a constant value of the lithium-ion transference number was produced across the spatial extent of the cell. A follow-up study combining MR images of the concentration gradients and ex situ PFG-NMR and ionic conductivity measurements verified that the steady-state concentration gradients measured with the in situ MRI technique are consistent with those predicted by the Newman electrochemical model.[153] Bazak et al. then extended the use of the coupled PFG-MRI technique to a range of operating temperatures and applied current densities and demonstrated that, at low temperatures, it is actually mass transport limitations on the anodic side of the electrolyte which limit achievable current density, as opposed to depletion of cations on the cathodic side.[91] This spatially resolved extension of PFG-NMR in the battery electrolyte domain therefore furnishes two useful directions for further development of battery technology: simultaneous, comprehensive electrolyte characterization at a range of concentrations and direct validation of electrochemical models which are crucial to modern battery management systems.[91,147,151,152]

■ CONCLUSIONS

We have provided an introduction to basic PFG-NMR experimental parameters, along with appropriate set up methods for these and the artifacts which should be avoided for acquiring accurate diffusion coefficients under general conditions. In particular, we highlight the challenges in applying PFG-NMR to spin systems with short relaxation times (especially T_2), slower diffusion, and lower gyromagnetic ratio. Strong gradients are critical to overcoming these handicaps. When sample convection is occurring, primarily in bulk liquid samples, the apparent diffusion coefficient will exhibit artifacts which can be suppressed by the use of modified pulse sequences, physical barriers in the sample volume, and/or sample spinning. Internal gradient artifacts, usually a concern when conducting PFG-NMR measurements in the presence of heterogeneous interfaces, can be quite insidious and cannot be easily suppressed. Apparent diffusion coefficients in this case will differ for the same sample when measured in different external magnetic field strengths and using different PFG-NMR pulse sequences. It is therefore recommended that, for diffusion measurements at liquid/solid interfaces, experiments should be conducted under two different external magnetic fields or by using two different PFG-NMR pulse sequences to diagnose the presence of internal gradient artifacts.

Following this pedagogical overview and artifact survey, which we intend as a guide for new practitioners of the technique, we demonstrate many emerging uses of diffusion coefficient analysis through a review of selected literature examples, with an emphasis on applications in energy storage. In the solution state, PFG-NMR has been widely applied for extracting electrolyte properties such as transference number, solvation structure of active ions, and ion−ion/ion−solvent interactions between constituent molecules. We highlighted novel cases where PFG-NMR has been applied to the elucidation of structure/dynamics relationships beyond this conventional solution-state use case, with examples drawn from ionic liquids, highly concentrated electrolytes (e.g., WiSE, LHCE), polymer electrolytes, and solid-state ion conductors. Taken together, these examples demonstrate the considerable capability that PFG-NMR delivers to advanced electrolyte characterization. For completeness, we also briefly introduced other NMR techniques such as relaxation measurements and ODNP-NMR for use in measuring similar dynamics in systems which are not amenable to PFG-NMR.

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Notes

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